

Review

Smart Greenhouses in the Era of IoT and AI: A Comprehensive Review of AI Applications, Spectral Sensing, Multimodal Data Fusion, and Intelligent Systems

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Abstract

Smart greenhouses (SGHs) are controlled-environment agricultural systems that leverage digital technologies to optimize crop production and resource management. In particular, recent advances in artificial intelligence (AI) and the Internet of Things (IoT) have enabled the development of intelligent monitoring, predictive modeling, and automated decision-support systems within these environments. Against this backdrop, this comprehensive review synthesizes over 130 studies published between 2020 and 2025, with a focus on AI-driven monitoring, predictive modeling, and decision-support frameworks in SGH environments. More specifically, key application domains include microclimate regulation, crop growth assessment, disease and pest detection, yield estimation, and robotic harvesting. Moreover, particular attention is given to the interplay between AI methodologies and their data sources, encompassing IoT sensor networks, RGB, multispectral, and hyperspectral imaging, as well as multimodal data-fusion approaches. In addition, publicly available datasets, model architectures, and performance metrics are consolidated to support reproducibility and cross-study comparison. Nevertheless, persistent challenges are critically discussed, including data heterogeneity, limited model generalization across sites, interpretability constraints, and practical barriers to deployment. Finally, emerging research directions are identified, notably multimodal learning, edge-AI integration, standardized benchmarks, and scalable system architectures, with the overarching objective of guiding the development of robust, sustainable, and operationally feasible AI-enabled SGH systems.

Keywords: smart greenhouse; artificial intelligence; machine learning; deep learning; Internet of Things; multimodal data fusion; hyperspectral and multispectral imaging; crop monitoring



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1. Introduction

Ongoing climate change exerts growing pressure on the agriculture sector [1]. According to the World Meteorological Organization (WMO), the past eleven years from 2015 to 2025 present the warmest period on record, as well as the temperature from January to August 2025 was approximately $1.42\text{ }^{\circ}\text{C} \pm 0.12\text{ }^{\circ}\text{C}$ above pre-industrial levels [2,3].

In parallel, the frequency and intensity of extreme events, including heatwaves, droughts, and heavy rainfall, have increased, disrupting crop schedules and exacerbating yield instability [4]. Over the past five decades, heat and drought have reduced global yields of key crops such as maize, wheat, and barley by 4–13% [5].

Furthermore, abiotic stressors have been shown to exacerbate biotic risks; Heat and water deficits can accelerate pest and pathogen life cycles and broaden their geographic ranges, thereby amplifying crop losses [6]. The FAO (2021) estimates that pests and diseases account for about 40% of global crop losses annually, resulting in economic losses of approximately 220 billion USD [7]. A 2025 review reported that warming of around 2 °C may increase pest-related yield damage by up to 46% in wheat and by more than 30% in maize [6].

Moreover, farming labor availability is declining due to urbanization, coupled with an aging population and a reduced participation of young workers as well [8,9]. For example, in Europe, less than 5% of workers are employed in agriculture, and globally, the proportion of young workers in agrifood systems in most countries has declined by around 10% in 2023 [9,10]. Meanwhile, projections indicate that the global population will peak at roughly 10.3 billion in the mid-2080s [11].

The aforementioned challenges highlight the vulnerability of the conventional open-field agricultural sector in the face of climate change, pest and disease pressure, and labor shortages [1,7,10]. Consequently, many sustainable strategies increasingly aim to ensure stable yield and quality under resource constraints and climatic uncertainty [12]. Thus, controlled-environment agriculture (CEA), specifically SGHs, offers a practical route to mitigate these challenges [12,13].

On the one hand, CEA is being more widely adopted, especially in developed countries, and serves as a strategy to support open-field agriculture and mitigate its vulnerability to climatic stress [13,14]. Previous studies highlight the capacity of CEA to deliver more consistent yields under climatic unpredictability [13], and describe CEA as an important approach for enhancing food security [12,13]. On the other hand, the integration of information and communication technologies (ICT) has become a predominant means in modern agriculture [15]. SGHs have appeared as an innovative technological solution that promises efficiency, precision, and sustainability [16].

Beyond that, recent advances in IoT-enabled sensing, high-resolution imaging, GPU-accelerated computing, and artificial intelligence (AI), including machine learning (ML) and deep learning (DL), have catalyzed the evolution of SGHs, where continuous data acquisition is coupled with data-driven analytics and automated actuation. In this paradigm, SGHs enable continuous monitoring of crop growth and health, regulation of internal variables such as humidity, temperature, CO₂, and light intensity, and robotic harvesting [16,17]. Figure 1 presents a conceptual overview of the main components of AI-driven smart greenhouse systems discussed in this review, including sensing, processing, and application domains.

Numerous studies have investigated the integration of AI and IoT techniques in smart greenhouses (SGHs). Escamilla-García et al. (2020) [18] reviewed artificial neural network (ANN) applications for microclimate prediction, energy optimization, and plant growth modeling. Liu et al. (2024) [19] focused on machine learning-based time-series forecasting of greenhouse environmental parameters. Akbar et al. (2024) [20] surveyed deep learning-based computer vision techniques for greenhouse agriculture. Yu et al. (2025) [21] reviewed AI- and IoT-enabled sensing technologies for tomato production. Al-Najadi et al. (2025) [16] examined the integration of IoT, AI, and robotics in autonomous greenhouse systems, while Al-Qudah et al. (2025) [22] surveyed AI applications and sensory data

utilization in SGHs for sustainable agriculture. The most relevant surveys are summarized in Table 1; further details are presented in Table A1 provided in Appendix A.

Although existing surveys provide valuable perspectives, they often focus on a single task (e.g., microclimate prediction), a specific technology (e.g., vision-based DL), or a particular crop. A consolidated, data-centric synthesis that jointly examines (i) multi-sensor data acquisition, (ii) multimodal data fusion across IoT and imaging modalities (RGB/MSI/HSI), and (iii) AI models for the full range of SGH applications remains limited.

To address this gap, this review makes four contributions: (i) Comprehensive coverage (2020–2025): synthesis of 133 peer-reviewed studies spanning microclimate prediction, crop growth and yield estimation, pest and disease monitoring, and robotic harvesting; (ii) model-centric analysis: structured comparison of single, hybrid, and ensemble ML/DL architectures across sensing modalities; (iii) reproducibility support: consolidation of publicly available datasets and commonly used evaluation metrics; and (iv) critical outlook: discussion of deployment barriers, generalization challenges, and future research directions for scalable and sustainable SGH systems.

The remainder of this paper is organized as follows: Section 2 describes the PRISMA-based review methodology and study selection process. Section 3 reviews the major SGH application domains covered in this review, including microclimate modeling, crop health monitoring and diagnosis, crop growth, yield estimation, and robotics harvesting, and consolidates the evaluation metrics used in the reviewed studies. Section 4 synthesizes HSI and MSI imaging in SGHs, Section 5 emphasizes multi-data source and multimodal data fusion in SGHs, Section 6 is dedicated to work incorporating AI and IoT in SGHs, Sections 7 and 8 discuss recurring challenges and future research directions, Section 9 presents the discussion, and Section 10 concludes the review.

A Conceptual Overview of AI-Driven Smart Greenhouse Systems

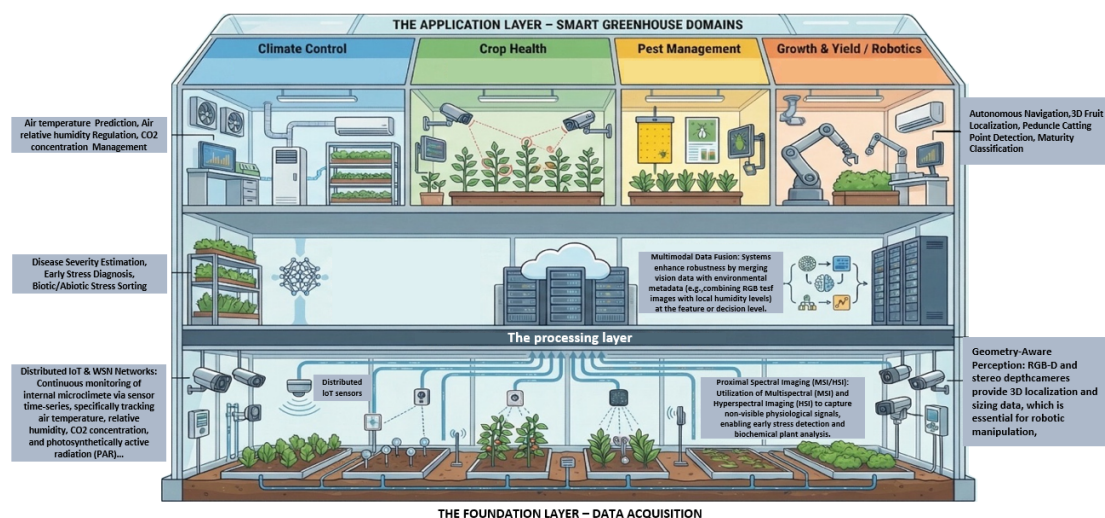


Figure 1. Conceptual overview of AI-driven smart greenhouse systems, highlighting the main sensing, processing, and application components discussed in this review.

Table 1. Compact summary of related survey papers on smart greenhouse (SGH) systems.

Ref.	Period	Papers	Domain	Tasks	Summary
[16]	2021–2025	100	Intelligent and autonomous greenhouse systems	Sensing; prediction/control; disease detection; growth/yield assessment; irrigation/fertilization; robotics	Reviewed AI, IoT, ML/DL, and robotics for autonomous greenhouse management. It concludes that multimodal fusion and IoT improve monitoring and automation, while robotics can reduce labor demands, but standardization, interpretability, generalization, and deployment cost remain key barriers.
[18]	NR	NR	ANN applications in greenhouse technology	Prediction/control; energy optimization; CO ₂ management	Examined ANN use in greenhouse systems and highlights that feedforward ANNs dominate, while recurrent and hybrid ANNs are less explored. Optimization methods such as GA, PSO, fuzzy logic and PCA improve ANN training and performance. The ANN models can surpass physical models for short-term prediction.
[19]	NR	61	ML/DL time-series forecasting in greenhouse environments	Microclimate prediction; water quality and dissolved oxygen forecasting; energy-related variables	Mapped the development of time-series forecasting models in greenhouse-related environments and finds that deep learning outperforms traditional methods for nonlinear spatio-temporal data. Attention and memory-based architectures improve long-sequence prediction, while shorter sampling intervals increase accuracy.
[20]	2020–2023	100	Deep-learning-based computer vision in greenhouse applications	Growth monitoring; disease detection; yield estimation; classification; harvesting; pest and weed monitoring	Reviews DL-enabled computer vision for greenhouse agriculture and shows strong potential for automation. However, progress is limited by scarce labeled datasets, weak transferability from open-field data, insufficient lightweight models for edge deployment, and underused multimodal sensing.
[21]	NR	NR	Sensing and AI for tomato greenhouse production	Environmental and physiological sensing; disease/stress detection; yield/quality assessment; irrigation/fertilization management	Examined non-invasive sensing, multi-sensor fusion, and AI diagnostics for tomato greenhouse production. It found that multimodal deep learning improves disease and stress diagnosis, early sensing supports proactive management, and IoT/WSN integration enables more precise automated control.
[22]	2020–2024	74	AI-driven smart greenhouses	Climate control; secure data handling; disease/pest detection; growth evaluation; big data management	Provides a state-of-the-art synthesis of AI and sensing in SGHs and identifies persistent limitations in multimodal datasets, benchmarks, and real-time deployment. ML/DL dominate the field, while RL, LLMs, XAI, and generative AI are still weakly represented.
[23]	2019–2020	73	mechanistic, time-series, and ML approaches	Microclimate prediction; climate control and optimization	Reviewed greenhouse modeling approaches and showed that dynamic and ANN-based methods dominate. RNN/LSTM adoption increased during 2018–2020, heuristic/swarm optimization is widely used, and temperature and humidity are the most common prediction targets.
[24]	2014–2023	107	Environmental control strategies and greenhouse models	Heating; cooling; lighting; ventilation control; mechanistic and data-driven modeling	Reviewed greenhouse control strategies and modeling methods, showing that PID remains common while MPC and neural-network control are increasingly adopted. It also notes frequent use of heuristic optimizers and continued emphasis on temperature and humidity regulation.
[25]	2010–2025	106	Conventional/intelligent control and modeling techniques	Climate control; irrigation; crop-growth control; NN- and RL-based controllers	Systematically summarized conventional and intelligent control approaches together with modeling techniques. It highlights the growing role of neural networks and reinforcement learning while pointing out gaps in real-time adaptability, sustainability, and integrated control model design.

NR: Not reported.

2. Literature Search Methodology

This literature review was conducted in accordance with the PRISMA guidelines. The review process consisted of three main phases: identification, screening, and eligibility, within the present research.

2.1. Identification and Selection Criteria

To identify recent and relevant research on SGHs, we performed a structured literature search in various well-established international academic databases, including Google Scholar, Scopus, IEEE Xplore, Springer, MDPI, and ScienceDirect.

The search strategy was developed using well-defined search strings to capture a broad range of studies relevant to SGHs. In the first stage, each of these following keywords: “disease detection,” “pest detection,” “insect detection,” “climate control,” “weed detection,” “hyperspectral imaging,” “multispectral imaging,” “synthetic images,” “artificial intelligence,” “machine learning,” “deep learning,” “computer vision,” and “multimodal fusion” were combined with “smart greenhouses” using the boolean operator AND (e.g., “disease detection” AND “smart greenhouses”). To broaden the search and capture additional relevant studies, we grouped semantically related keywords with the OR operator and linked these groups with AND. This procedure yielded search queries, for example, for the form: “synthetic images” AND “smart greenhouses” AND (“disease detection” OR “weed detection”) AND (“deep learning” OR “computer vision”).

Selection criteria were then incorporated to determine which articles should be included or excluded. Studies were selected to be included in this review based on their publication years, so an eligible article had to be published in the last five years (2020–2025), written in English, and published in peer-reviewed journals.

Only advanced SGH applications were considered. In particular, we included research that reported concrete implementations of technologies such as AI, ML, DL, and IoT. The selected articles were required to address at least one of the following topics: disease detection, pest/insect detection, climate prediction, robotic harvesting, yield estimation, crop growth, HSI, MSI, and data fusion.

Whereas articles were excluded if they focused exclusively on open field agriculture or other controlled environments rather than SGHs, if they were not written in English or if they were not peer-reviewed articles.

2.2. Screening

Upon removing duplicate studies from the search results, the screening process was carried out based on two stages; the initial stage involved an examination of the titles and the abstracts, followed by the second stage, where the full-texts were examined to confirm the selection of studies. The quality of each selected article is evaluated using the inclusion and exclusion criteria.

2.3. Results

The digital search libraries returned around 327 records in the initial search, including $n = 200$ from Google Scholar, $n = 20$ from Scopus, $n = 10$ from IEEE Xplore, $n = 6$ from Springer, $n = 30$ from ScienceDirect, and $n = 61$ from MDPI.

The first screening resulted in removing duplicates $n = 120$, then titles and abstracts were screened to exclude clearly irrelevant studies, resulting in the elimination of 60 studies at this stage. Finally, the full texts of the remaining 147 articles were then assessed for eligibility based on the predefined inclusion and exclusion criteria. Finally, 133 research articles published between 2020 and 2025 were selected for the present review. An overview of the flowchart following PRISMA guidelines is presented in Figure 2.

The articles reviewed in this study are on the accompanying GitHub repository (https://github.com/WiamElouaham1998/SGHs_review_studies, accessed on 15 February 2026), which is actively maintained. The selected papers were classified into different domains such as objective of the study, climate control, disease, pest and insect detection, weed management, growth evaluation, or yield estimation. Figure 3 illustrates the percentages of each application type among the included papers.

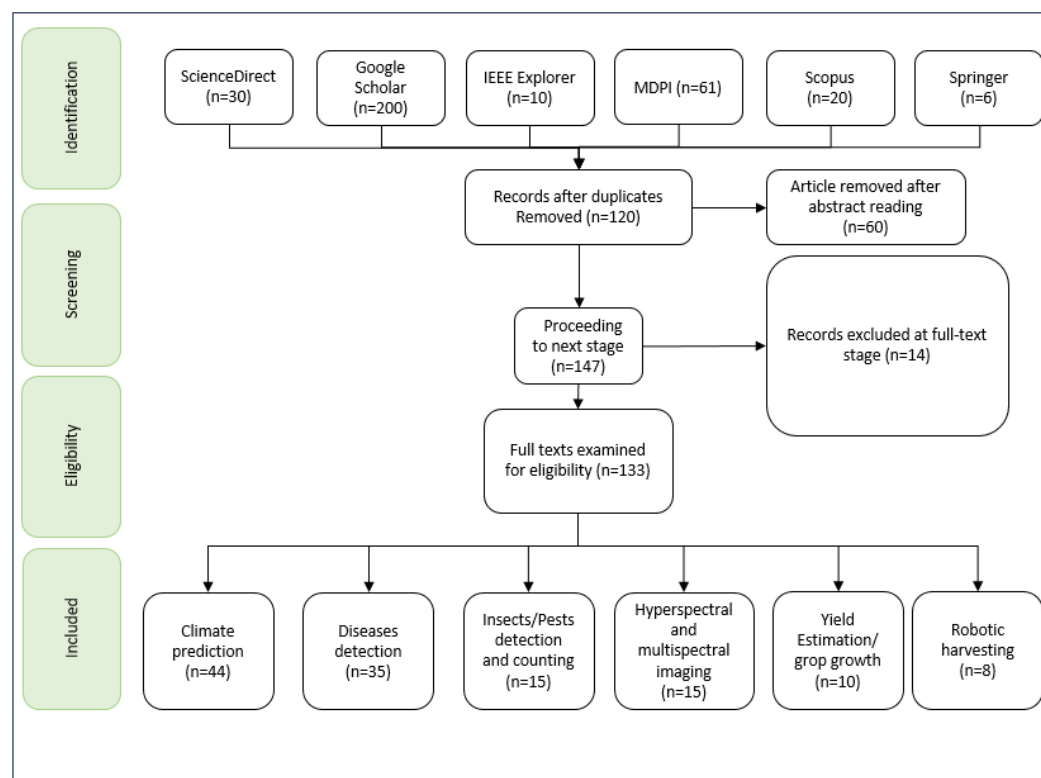


Figure 2. PRISMA diagram of the study selection process.

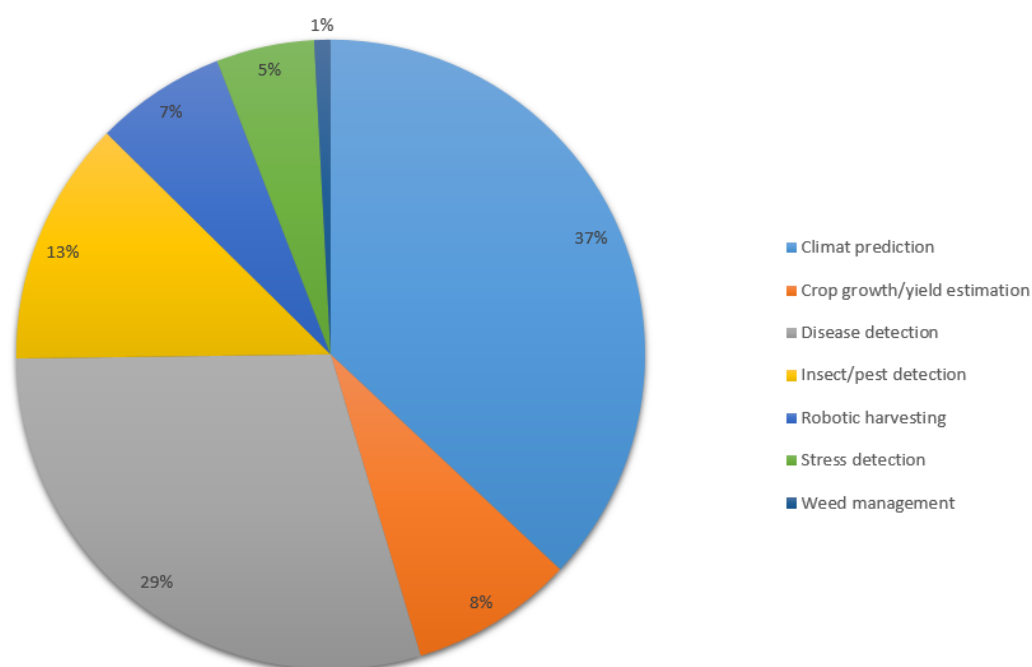


Figure 3. Distribution of included studies across application domains.

3. AI Applications in SGHs

3.1. Microclimate Prediction and Climate Control in SGHs

Microclimate prediction in SGHs is most often formulated as a time-series regression (forecasting) problem, where key indoor variables such as air temperature, relative humidity, and CO₂ concentration are forecast from historical sensor data. Accordingly, the literature has investigated a broad spectrum of data-driven models, ranging from classical statistical formulations to machine-learning predictors and, more recently, deep sequence architectures for multi-step forecasting.

In early work, statistical time-series techniques were extensively adopted as competitive baselines, particularly for temperature prediction, including autoregressive with exogenous input (ARX), autoregressive integrated moving average (ARIMA), and seasonal autoregressive integrated moving average (SARIMA) [26–28]. For instance, Frausto et al. [26] compared ARX and ARMAX for greenhouse air-temperature prediction and reported that ARX achieved a higher coefficient of determination $R^2 = 0.965$ than ARMAX $R^2 = 0.888$. Nevertheless, despite their effectiveness in capturing linear trends in historical data, purely statistical models often exhibit limited capacity to represent the nonlinear, coupled, and regime-dependent dynamics that characterize greenhouse microclimates [23,29].

Classical machine learning methods and neural network predictors have been increasingly employed to model complex nonlinear relationships and to exploit larger datasets [30,31]. In this vein, García-Vázquez et al. [32] proposed support vector regression (SVR) for temperature forecasting and demonstrated that a polynomial-kernel SVR reduced error relative to linear baselines, achieving in summer an RMSE of 0.0549 °C and an MAE of 0.0422 °C, outperforming PLS-based regression, OLS multiple linear regression, and an SVR with an RBF kernel. Similarly, shallow neural predictors, including LM-trained RBF networks, have been repeatedly investigated to improve short-term temperature forecasts by incorporating exogenous drivers [33,34]. Beyond temperature, NARX-type models have also been explored for humidity forecasting; moreover, comparisons among LM, BR, and SCG training procedures indicate that optimization choices can materially affect convergence and generalization [35].

Importantly, the transition from one-step prediction to multi-step forecasting is more directly aligned with scheduling and closed-loop control objectives. However, the key bottleneck is capturing longer temporal dependencies under non-stationarity. Therefore, recent studies have increasingly evaluated recurrent architectures while explicitly acknowledging deployment constraints. For example, Codeluppi et al. [36] compared ANN, RNN, and LSTM models for greenhouse air temperature forecasting and underscored that operational utility depends not only on accuracy but also on computational feasibility for IoT/edge execution. In the same context, GRU-based predictors have been reported to outperform classical machine learning, including random forest (RF), support vector machine (SVM) and multiple linear regression (MLR) for temperature forecasting [37], supporting the recurring claim that gated recurrence is better suited to delayed and regime-dependent greenhouse dynamics, which further supports the view that gated recurrence is well-suited to delayed responses and regime-dependent greenhouse dynamics. For longer horizons and sequence-to-sequence configurations, attention mechanisms are frequently introduced to alleviate the encoder–decoder information bottleneck and to selectively emphasize relevant historical information. For instance, Luong-attention-based seq2seq models have been explored for temperature and humidity forecasting [38], whereas feed-forward attention mechanisms combined with LSTM, reported as FAM-LSTM, have been reported to maintain competitive accuracy across horizons up to 48 h in solar greenhouses [39]. Taken together, existing evidence suggests a horizon-dependent modeling strategy: gated recurrent models are often adequate for moderate horizons, while attention-enhanced sequence modeling

becomes increasingly advantageous as the forecast horizon lengthens and the relevant history expands.

Furthermore, ensemble learning has been adopted to improve accuracy and, in some cases, reduce the cost of retraining; notably, LightGBM has been reported to provide strong predictive performance while substantially reducing training time when compared with baseline models such as Back-Propagation (BP) neural networks, recurrent neural network (RNN) models, Extreme Gradient Boosting (XGBoost), and Stochastic Gradient Boosting (SGB) [40]. In parallel, hybrid approaches go further by coupling mechanistic structure with data-driven correction. A representative pipeline combines a process-based greenhouse climate–crop model built from GreenLight and TOMSIM with particle-filter calibration, then applies a DNN correction stage to further reduce forecast error [41–44], aiming to preserve physical plausibility while absorbing unmodeled nonlinearities. Complementary multimodel combinations, including DE, RF, and ANN, have also been reported to enhance multivariable microclimate prediction [45]. Moreover, Variable-weight ensembles optimized by PSO further aim to adaptively fuse predictors across horizons from 30 to 120 min [46]. Table 2 synthesizes the reviewed studies and reports additional methodological and experimental details; further details are presented in Table A2 provided in Appendix A.

Table 2. Summary of data-driven microclimate forecasting studies in SGHs.

Ref./Year	Crop	Category	Architecture	Performance	Target	Interval	Dataset	Period	Dataset Availability
[29] (2023)	NR	Hybrid modeling	ARTFIMA-SVM	Air temperature: RMSE = 0.79, MSE = 0.62, MAE = 0.44, MAPE Moisture: RMSE = 1.94, MSE = 3.79, MAE = 1.46, MAPE = 0.02	Internal air temperature, Moisture	10 min	1987 samples	NR	Private
[32] (2023)	Tomato, bell pepper	Classical machine learning	SVR + polynomial kernel	Fall: $R^2 = 0.9808$, RMSE = 1.7431, MAE = 0.2856, MAPE = 0.0124, Winter: $R^2 = 0.9984$, RMSE = 0.6760, MAE = 0.2929, MAPE = 0.0123, Spring: $R^2 = 0.9924$, RMSE = 1.0362, MAE = 0.7444, MAPE = 0.0270, Summer: $R^2 = 0.9999$, RMSE = 0.0549, MAE = 0.04222, MAPE = 0.0015	Internal air temperature	5 min	85,989 samples	2020–2021	Public
[33] (2023)	Cucumber	Classical machine learning	RBF	RMSE = 1.32 C, MAPE = 3.23%, $R^2 = 0.931$	Internal air temperature	5 min	NR	1 month (2020)	On request
[34] (2023)	NR	Classical machine learning	RBF + Levenberg–Marquardt	RMSE = 0.91, MAPE = 1.30	Internal air temperature	5 min	NR	2022 (10 days)	On request
[36] (2021)	NR	Shallow Neural Networks	ANN	RMSE = 0.289–0.402, MAPE = 0.87–1.04%, $R^2 \geq 0.997$	Internal air temperature	10 min	40,033 samples	08/2019–11/2020	On request
[37] (2022)	NR	Deep sequential learning	GRU	$R^2 = 0.87$, RMSE = 3.58	Minimum Internal air temperature	1 h	NR	20/2017–30/2018	On request
[38] (2023)	NR	Deep sequential learning	Seq2seq GRU with Luong attention	Air temperature: MAE = 0.228760, RMSE = 0.274486, $R^2 = 0.626864$. Realtive humidity: MAE = 1.420797, RMSE = 1.666903, $R^2 = 0.843962$	Internal Air temperature, Internal Air relative humidity	NR	273,144 timesteps	NR	On request
[39] (2023)	Cucumber	Deep sequential learning	FAM-LSTM	MAE 0.6 °C, RMSE 0.9 °C (12 h); MAE 0.8 °C, RMSE 1.1 °C (24 h); MAE 1.4 °C, RMSE 1.7 °C (36 h); MAE 1.7 °C, RMSE 2.3 °C (48 h). Humidity (split 8:2): MAE 0.8%, RMSE 1.0% (12 h); MAE 2.5%, RMSE 3.1% (24 h); MAE 2.6%, RMSE 3.7% (36 h); MAE 2.5%, RMSE 3.5% (48 h)	Internal air temperature, Air relative humidity	5 min	23,344 samples	2021–2022	On request
[40] (2022)	Cucumber	Ensemble learning (Boosting)	LightGBM (LGBM)	RMSE = 0.645	Internal air temperature	5 min	495,714 samples	2014–2019 (2016 excluded)	Public
[41] (2023)	Tomato	Hybrid modeling	PBI + PF + DNN	temperature: RRMSE 6.7%, vapor-pressure: RRMSE 12.6%; seasonal total fresh-yield error 0.9%, cumulative-yield RRMSE 6.6%.	Internal air temperature, Internal vapor pressure, Seasonal fresh yield prediction	5 min	NR	2015–2020	Private
[45] (2023)	Cherry tomato	Hybrid modeling	DF-RF-ANN	Air temperature: $R^2 = 0.72$, MAE = 1.97, Air relative humidity: $R^2 = 0.68$, MAE = 0.10, PAR: $R^2 = 0.75$, MAE = 76.39, CO ₂ concentration: $R^2 = 0.59$, MAE = 10.14,	Internal air temperature, Air relative humidity, Photosynthetically active radiation (PAR), CO ₂ concentration	10 min	NR	01/04/2020–13/07/2021	On request

Table 2. Cont.

Ref./Year	Crop	Category	Architecture	Performance	Target	Interval	Dataset	Period	Dataset Availability
[46] (2024)	Cherry tomato	Hybrid ensemble learning	PSO-BiGRU-Attention-LightGBM	30 min: Air Temperature $R^2 = 0.9902$ RMSE = 0.5578, MAE = 0.3172; Air relative humidity $R^2 = 0.9825$, RMSE = 2.2327, MAE = 1.1981; PAR $R^2 = 0.8871$, RMSE = 39.1562, MAE = 18.4348. 60 min: Air Temperature RMSE = 0.7936, MAE = 0.4793, $R^2 = 0.9803$; PAR: RMSE = 46.2710, MAE = 22.8484, $R^2 = 0.8423$; Air relative humidity: RMSE = 3.1482, MAE = 1.7270, $R^2 = 0.9652$, 120 min: Air Temperature $R^2 = 0.9586$; Air relative humidity $R^2 = 0.9232$; PAR $R^2 = 0.8066$.	Air temperature, Air relative humidity, Photosynthetically active radiation PAR	10 min	37,008 samples	23/09/2020–06/06/2021	On request
[47] (2023)	Tomato	Deep sequential learning	LSTM-RNN	$R^2 = 0.80$, RMSE = 9.13%, MAE = 4.45%, MaxAE = 62.94%	Vent opening	30 s	233,280 samples	10/12/2020–29/12/2020	On request
[48] (2024)	Ornamental plants	Ensemble learning (Stacking)	Stacking DT + RF + KNN + XGB	$R^2 = 0.96515$, MAE = 0.01395, MSE = 0.03205, RMSE = 0.00102.	Internal air relative humidity	15 s	604,135 samples	08/2021–12/2021	On request
[49] (2023)	NR	Deep sequential learning/ensemble learning (boosting)	LSTM-RNN, XGBoost	LSTM-RNN best result in summer: $R^2 = 0.9994$, RMSE = 0.2698, MAE = 0.1449, MAPE = 0.0041; XGBoost best result in winter: $R^2 = 0.9345$, RMSE = 4.4443, MAE = 2.7693, MAPE = 0.0809	Air temperature	NR	NR	07/2020–06/2021	Public
[50] (2025)	Tomato	Deep sequential learning	PLSTM	$R^2 = 0.9999$, RMSE = 0.0220, MAE = 0.0160	Internal air temperature	NR	NR	15/02/2018–14/08/2018	On request
[51] (2025)	NR	Hybrid modeling	LSTM-SVM	Air temperature (LSTM best): RMSE = 0.0766, MAE = 0.0454, $R^2 = 0.8825$; Air relative humidity (best overall): RMSE = 5.3034, MAE = 3.8041, $R^2 = 0.8187$; Classification (SVM): Accuracy = 0.63, Precision = 0.64, Recall = 0.63, F1-score = 0.63;	Air temperature, Relative air humidity, Environmental states	10 min	NR	NR	Private
[52] (2024)	Tomato	Deep sequential learning	DLinear, SegRNN	1 h—DLinear (Air temperature): $R^2 = 0.938$, RMSE = 0.293, MAE = 0.189; (Air relative humidity): $R^2 = 0.857$, RMSE = 0.427, MAE = 0.273; SegRNN (CO ₂ concentration): $R^2 = 0.875$, RMSE = 0.371, MAE = 0.231. 3 h—DLinear: (Air temperature): $R^2 = 0.833$, RMSE = 0.479, MAE = 0.312; (Air relative humidity): $R^2 = 0.680$, RMSE = 0.640, MAE = 0.458; SegRNN (CO ₂ concentration): $R^2 = 0.711$, RMSE = 0.552, MAE = 0.354	Internal air temperature, Internal air relative humidity, CO ₂ concentration	NR	6744 samples	22/09/2020–29/06/2021	On request
[53] (2025)	Tomato	Hybrid modeling	GWO-BiLSTM	10 min: $R^2 = 0.97$, RMSE = 0.7889, MAPE = 4.94, MSE = 0.63, MAE = 0.62; 30 min: $R^2 = 0.97$, RMSE = 0.8863, MAPE = 8.5, MSE = 0.68, MAE = 0.65	Internal air temperature	5 min	124,837 samples	13/09/2024–1/11/2024	On request
[54] (2025)	NR	Classical machine learning	RF, MLR	Sunny: RF: $R^2 = 0.78$, RMSE 1.37 °C, MAPE 4.54%, Cloudy: RF: $R^2 = 0.81$, RMSE 1.33 °C, MAPE 4.14%, Overcast: MLR: $R^2 = 0.82$ (five-fold CV RMSE 0.68 °C); RF lower at $R^2 = 0.72$.	Internal air temperature	1 h	NR	11/2024–01/2025+2021–2025	Private
[55] (2025)	NR	Hybrid modeling	RIME-CNN-BiLSTM	Air temperature: MAE = 0.598, RMSE = 0.826, $R^2 = 0.977$. Air relative humidity: MAE = 1.698, RMSE = 2.287, $R^2 = 0.981$	Internal air temperature, Internal air relative humidity	15 min	35,040 samples	23/07/2020–22/07/2021	Private

Table 2. Cont.

Ref./Year	Crop	Category	Architecture	Performance	Target	Interval	Dataset	Period	Dataset Availability
[56] (2025)	Celery	Classical machine learning	RBF, MLP trained + Levenberg–Marquardt (LM)	30 min: Air temperature: RMSE = 1.579 °C, $R^2 = 0.958$ (RBF). 30 min: Air relative humidity: RMSE = 4.299%RH, $R^2 = 0.948$ (MLP). Current-time prediction: Air temperature RMSE = 0.439 °C, $R^2 = 0.997$ (MLP); Air relative humidity RMSE = 1.141%RH, $R^2 = 0.996$ (MLP)	Internal air temperature, Internal air relative humidity	10 min	8441 samples	22/12/2021–19/02/2022	Public
[57] (2022)	NR	Deep sequential learning	DNNR, LSTM,CNN-LSTM	DNNR was the best overall model for 3 h, 6 h, and 24 h ahead prediction, while LSTM/CNN-LSTM performed better at 12 h ahead. Reported best-model error ranges: Air relative humidity (DNNR)—3 h: [−8, 10], 6 h: [−12, 13], 24 h: [−19, 26]; Air temperature (DNNR)—3 h: [−5, 8] °C, 6 h: [−6, 6] °C, 24 h: [−7, 7] °C. For 12 h ahead, LSTM/CNN-LSTM had smaller errors than DNNR	Internal air temperature, Internal air relative humidity	1 min	1073 samples	GH 1: 1/05/2018–20/07/2018, GH 2: 01/05/2020–20/06/2020	On request
[58] (2025)	Cucumber, Melon	Hybrid modeling	Transformer + BiLSTM	MSE = 0.0267 °C, $R^2 = 0.9995$	Internal air temperature	5 min	NR	GH 1: 01–03/2024, GH 2: 12–03/2024	Private
[59] (2024)	NR	Hybrid modeling	GCAKF-CNN-LSTM	Air temperature: RMSE = 0.065 °C, MAE = 0.032 °C, Air relative humidity: RMSE = 1.103%, MAE = 0.346%	Internal air temperature, Internal air relative humidity	1 min	25,497 samples	14/03–03/04 2019	NR
[60] (2025)	Tomato seedlings	Deep sequential learning	LSTM-AT-DP	Air temperature: $R^2 = 0.9602$, MAE = 1.4930 °C, RMSE = 1.6843 °C. Air relative humidity: $R^2 = 0.9529$, MAE = 3.1278%, RMSE = 3.9557%. Solar radiation: $R^2 = 0.9839$, MAE = 12.5898 W/m ² , RMSE = 19.5745 W/m ²	Air temperature, Air relative humidity, Solar radiation	15 min	NR	04–08/2024 06/03–26/06/2025	On request
[61] (2025)	NR	Hybrid modeling	BO-LSTM	Air temperature RMSE = 0.626 °C, $R^2 = 0.986$; Air relative humidity (absolute) RMSE = 0.309 g/m ³ , $R^2 = 0.987$	NR	Air temperature, Absolute humidity	5 min	29/12/2021–10/04/2022 3–10 04/2022, 21–29 12/2022, 1–8 02/2023	Private
[62] (2024)	Chili pepper seedlings, tomato seedlings	Hybrid modeling	CNN-SE-LSTM	Air temperature: MAE = 0.540 °C, RMSE = 0.755, $R^2 = 0.940$, Air relative humidity: MAE = 0.936%, RMSE = 1.618, $R^2 = 0.951$, Solar radiation: MAE = 1.586 W/m ² , RMSE = 3.417, $R^2 = 0.936$	Air temperature, Air relative humidity, Solar radiation	10 min	13,632 samples	10/04–30/08 2023	On request
[63] (2024)	Tomato	Hybrid modeling	LSTM2	$R^2 = 0.962$, MAPE = 3.216%, RMSE = 1.196 °C	Air temperature	10 min	127,623 samples	GH1: 6/11/2019–08/12/2021, GH2: 1/10/2021–31/01/2022	On request
[64] (2024)	NR	Deep sequential learning	GRU	Average $R^2 = 0.8811$, RMSE = 2.056 °C	Air temperature	1 min	NR	07/10/2021–05/11/2021	Public

NR: Not reported.

3.2. Disease Detection in SGHs

Crop disease detection represents a challenging task for achieving an optimal crop yield production, as outbreaks may progress rapidly once initiated. Traditional manual crop disease monitoring is typically slow and subjective, whereas laboratory-based crop disease monitoring provides objective results but is often time-consuming and costly. Due to these limitations, AI-based approaches have been increasingly applied to crop disease detection. Early methods utilized traditional ML based on handcrafted features, while the recent DL-based approaches enable more automated disease detection. In RGB-based detection methods, transfer learning has been widely adopted to improve the model accuracy and precision, particularly when only limited labeled data are available. For instance, Zhang et al. [65] fine-tuned EfficientNetB0–B7 for multiclass cucumber leaf disease classification and reported optimizer-driven improvements for separating difficult classes. Related work has also explored multi-task transfer learning using EfficientNetB4 across rust diseases and multiple crops [66].

For disease localization, real-time detectors are commonly strengthened through lightweight backbones and improved box regression. In this context, MobileNetV2-YOLOv3 replaces the Darknet-53 backbone to reduce computation while enhancing localization via GIoU loss, and it preserves multi-scale detection to better capture small lesions in complex greenhouse scenes [67]. Precision under small, occluded, and multi-scale symptoms has been further improved by redesigning the feature reuse and training strategy. For example, YOLO-Dense, which substitutes dense connections for residual blocks, refines anchors with improved K-means clustering, and adopts multi-scale training to raise mAP while maintaining low latency [68]. Robust recognition in cluttered backgrounds has also been addressed through attention-guided region focusing and progressive learning. PRP-Net introduces an attention proposal mechanism to localize discriminative lesion regions without manual bounding boxes and employs channel attention to emphasize symptom-relevant features [69]. Similarly, SE attention has been integrated into YOLOv5 backbones to amplify disease symptoms and suppress irrelevant background information while maintaining real-time feasibility [70].

In strawberry disease monitoring, deployment-oriented designs explicitly trade architecture efficiency for edge readiness, with DAC-YOLOv4 compressing the backbone and neck using depthwise blocks, adding CBAM, re-estimating anchors with K-means, and adopting CIoU loss and DIoU-NMS to support real-time inference on Jetson devices [71]; two-stage pipelines also benefit from multi-scale feature fusion and attention, where a ResNet-FPN backbone augmented with CBAM and a dedicated multi-scale feature fusion module improves detection under greenhouse background complexity [72]. A lightweight YOLOv5 variants further demonstrate that accuracy and speed can be improved simultaneously by combining efficient operators and attention, as in YOLO-GIC-C, which integrates GhostConv, involution, CBAM, and CARAFE upsampling to reduce parameters and FLOPs while increasing mAP and FPS [73].

Beyond detection and classification, severity estimation is increasingly treated as a decision-relevant output. Accordingly, two-stage designs have been proposed that first identify diseased leaves and then segment diseased tissue using a U-Net-style model, thereby computing severity as a diseased-area ratio [74]. Table 3 synthesizes the RGB-based disease detection models in SGHs adopted by researchers; further details are presented in Table A3 provided in Appendix A.

Table 3. Summary of RGB-based disease detection in SGHs.

Ref./Year	Crop	Task	Architecture	Performance	Inference Time	Dataset	Dataset Availability
[65] (2020)	Cucumber leaf	Classification	EfficientNet-B4 + Ranger	Acc: 97%	NR	2816 images (resized 3000 × 3000) augmented to 20,259 (training 380 × 380)	Private
[66] (2023)	Sunflower, Dry bean, Field pea (leaves)	Detection	EfficientNetB4 + Adam optimizer	Ac: 95.56%, Pr: 96.86%, R: 93.65%, Sp: 97.25%, F1: 95.18%, AUC-ROC: 96.92%	NR	1764 images (resized to 224 × 224)	On request
[67] (2020)	Tomato leaf	Detection	MobileNetv2-YOLOv3	F1: 93.24%, AP: 91.32%, IoU: 86.98%	GPU: 246 fps/16.9 ms, CPU: 22 fps/80.9 ms	2385 images	NR
[68] (2021)	Tomato leaf	Detection	YOLO-Dense	mAP: 96.41%	20.28 ms	15,000 images (resized 544 × 544)	On request
[69] (2021)	Tomato, Cucumber (leaves)	classification	PRP-Net (ResNet18)	Acc: 98.26%, Pr: 92.60%, Sen: 93.60%, Sp: 99.01%	NR	4284 images (Resized to 448 × 448)	NR
[70] (2022)	Tomato leaf	Detection	SE-YOLOv5	Acc: 91.07%, mAP@0.5: 94.10%, P: 86.75%, R: 92.19%	50.63 ms (19.75 fps @ 640 × 640)	150 images (4032 × 3024), augmented to 1036 images (resized to 640 × 640)	NR
[71] (2022)	Strawberry leaf	Detection	DAC-YOLOv4	mAP@0.5: 72.7%, F1: 0.716, mp: 75.5%, R: 68.2%	43 FPS, 20 FPS	1023 images (1040 × 780, resized to 416 × 416), 764 augmented to 6112 training images	Public
[72] (2022)	Strawberry leaves, Flowers, Fruits	Detection	Multi-scale Feature Fusion Faster R-CNN (ResNet-50 backbone + FPN + CBAM)	mAP: 92.18%, AP (per class): 90.69–94.39%	229 ms	3600 images (2160 training, 720 validation augmented to 14,400)	NR
[73] (2023)	Strawberry (leaves, flowers, pedicels, and fruits)	Detection	YOLO-GIC-C (Improved YOLOv5s + Ghost-Conv + Involution + CARAFE + CBAM)	Pr: 93.3%, R: 90.3%, F1 ≈ 91.8%, mAP@0.5: 94.7%	92.6 FPS	Strawberry Disease Recognition 2246 images	Public
[74] (2020)	Tomato leaf	Severity estimation	U-Net + VGG16	MJ: (0.217–0.606), ME: 6.0% to 21.6%	NR	18,005 images (resized 256 × 256)	Public
[75] (2021)	Strawberry (leaves, flowers, pedicels, and fruits)	Segmentation	Mask R-CNN + ResNet101	mAP@0.5: 82.43%	NR	2500 images (resized to 419 × 419)	Public
[76] (2024)	14 Different crop	Classification	down-scaled dense residual connection	Acc: 96.75%, P: 97.62%, R: 97.59%, F1: 97.58%	31.7 ms	(1) Plant Village 54306 images (resized 256 × 256), (2) Strawberry Disease Recognition 2500 images (resized 256 × 256)	Public
[77] (2024)	Basil	Detection	YOLOv8 vs. YOLOv9	YOLOv8: P = 95.78%, R: 96.43%, F1: 0.96@0.544, mAP50: 97.39, mAP50–95: 87.22/YOLOv9: P: 95.93%, R: 96.64%, F1: 0.96@0.536, mAP50: 97.22, mAP50–95: 88.38	NR	4596 images	Public
[78] (2024)	Tomato, Cucumber, Eggplant	detection	YOLOv8n-vegetable	mAP@0.5: 92.91%, mAP@0.5: 0.95: 57.48%, P: 92.72%, R: 87.73%	271.07 fps	800 video sequences (40,000 keyframe images, Cropped and annotated to 28,000 images 640 × 640)	Partially public
[79] (2025)	Barley seedlings	Detection	YOLOv8-DDS	P: 98.9%, R: 96.6%, mAP@0.5: 98.5%	57.6 ms	1042 images (resized to 640 × 640, augmented to 4380)	Partially public
[80] (2025)	Cabbage, Shanghai Green, Napa Cabbage, Brassica Napus	Multi-class detection	IM-AlexNet	Pr: 88.61%, Rec: 81.46%, F1: 84.9%, mAP: 88.91%	NR	8000 images (720 × 1920 resized to 256 × 256)	On request

Table 3. Cont.

Ref./Year	Crop	Task	Architecture	Performance	Inference Time	Dataset	Dataset Availability
[81] (2025)	Tomato, Cucumber, Pepper	Detection	YOLO-vegetable (Improved YOLOv10n)	Pr: 95.8%, Rec: 94.9%, mAP@0.5 = 95.6%	18.6 ms/14.7 GFLOPs	15,000 images (resized to 640 × 640)	NR
[82] (2025)	Pepper	Detection	YOLO-Pepper (Based on YOLOv10n with AMSFE, DFPN, SDH, Inner-CIoU)	Pr: 94.69%, Rec: 93.87%, mAP@0.5: 94.26%	115.26 FPS	8046 images (1920 × 1080 resized to 640 × 640)	Public
[83] (2025)	Cucumber	Lesions detection	YOLO-Cucumber (baseline YOLOv11n) + C3k2-DCN + P2 small-target layer (remove P5) + TAL + CPBN pruning	mAP@0.5: 93.8%; mAP@0.5:0.95: 72%; P: 93%; R: 92%	218 FPS	4740 images (456 × 2304 pixels)	Partially public
[84] (2024)	Cucumber	Small-target detection	CucumberDet (RetinaNet + Swin Transformer backbone + SFFM + MFAFM (and removes P7 layer))	mAP: 92.5%; P: 92.9%; R: 92.0%; F1: 92.4%	23.6 FPS (paper also reports 72 FPS in a separate comparison table)	4740 images	Public
[85] (2024)	Tomato	Detection	TomatoDet (YOLOv8n + Swin-DDETR self-attention feature extraction module + Meta-ACON dynamic activation + IBiFPN multi-scale feature fusion)	mAP@0.5: 92.3%	46.6 FPS	2000 images (captured at 3648 × 2056) augmented to 9600 images	Partially public
[86] (2025)	Tomato (leaf)	Classification	TLDVLM: GroundingDINO (leaf detect) + SAM-2(segmentation) + BLIP-2 with LoRA on Q-Former	Acc 97.27%, P: 0.9587, R: 0.9789, F1: 0.9681	2.7 B (base) + 0.6 M (LoRA + classifier)	NR	NR
[87] (2021)	Strawberry	Detection	PlantNet (ResNet152 pre-trained on Plant-CLEF/LifeCLEF2017) + FPN, 2-stage cascade	mAP: 91.65%	0.662 s	4725 images (train + val 3309 augmented 39,708)	NR

NR: Not reported.

3.3. Pest and Insect Detection and Counting in SGHs

In smart greenhouse systems, computer-vision-based pest monitoring is increasingly used to replace manual scouting by converting image data into quantitative estimates of pest presence, species, and population density, thereby supporting timely interventions and reducing excessive pesticide application [88].

The literature generally categorizes pest detection according to acquisition scenarios, including insects on sticky traps, insects on plant surfaces, and insects on trap baseplates, as each scenario presents distinct detection challenges such as small target size, specular reflections, motion blur, and background complexity [88].

For sticky-trap monitoring, where small-object detection and counting are primary objectives, early approaches improved reliability under limited computational resources by combining saliency-based candidate extraction with color feature descriptors and SVM classification for pests such as whiteflies and thrips [89,90]. Recent deep learning models have further enhanced detection performance by explicitly addressing small-target characteristics. For example, TPest-RCNN improves anchor design and RoI feature alignment to preserve fine pest details, enabling robust detection across varying pest densities on reflective trap surfaces [91].

Deployment constraints have also driven the development of accuracy–efficiency trade-offs. Lightweight architectures such as SSD-Lite with a MobileNetV2 backbone enable real-time on-device inference on embedded platforms, including Raspberry Pi systems [92]. In parallel, resolution-preserving inference strategies based on high-resolution image tiling have been proposed to prevent the loss of small pest features during downsampling, as demonstrated by a cascade detection framework built on a tiny YOLOv3 model [93].

For on-plant pest detection, where visual variability and insect movement are more pronounced, deep learning pipelines have incorporated mobile-to-cloud architectures. Faster R-CNN-based systems have been used not only for pest recognition but also for linking detection outputs to decision-support recommendations for pest management [94]. In multi-species greenhouse environments characterized by dense clustering and occlusion, performance improvements have increasingly relied on data augmentation and post-processing techniques. Copy–paste augmentation enhances the representation of small pests in training datasets, while DIOU-based non-maximum suppression improves detection accuracy in crowded scenes, resulting in substantial performance gains over baseline YOLOv5 models [95].

Table 4 synthesizes the reviewed papers for insect and pest detection models in SGHs.

Table 4. Overview of insect and pest detection models in SGHs.

Ref./Year	Task	Crop	Image Scenario	Categories	Target	Model Type	Architecture	Performance	Inference Time	Model Size	Hardware	Camera	Dataset	Annotation Tool	Dataset Availability
[89] (2022)	Automatic detection & counting of small insects	Green pepper	Sticky traps	3	Adult-stage whitefly, Thrips, Background	Traditional ML	Spectral Residual saliency for candidate proposals + HSV color features + SVM (LIBSVM)	Whitefly: TPR 93.1%, FPR 9.9%, Acc 93.9%; Thrips: TPR 80.1%, FPR 12.3%, Acc 89.8%; Correlation vs. manual count: $R^2 = 0.9785$ (whitefly), 0.9572 (thrips)	NR	NR	NR	NR	Trap 25 × 20 cm: RGB images 2560 × 1920 every 2 h, 18 images (train) + 18 images (test); 1500 cropped 32 × 32 samples for training; test uses all targets in second set	Not specified	On request
[90] (2023)	Whitefly counting	NR	Full yellow chromotropic sticky traps	1	Count of whiteflies per image	Deep Learning	Faster R-CNN (ResNet-50)	MAE 65.75; MSE 8532.13; MARE 9.3%	NR	NR	NR	Nikon D5300	28 images; 20 train/8 test; 17,005 insects	LabelMe	Public
[91] (2021)	Tiny pest detection and counting	Green pepper	Yellow sticky traps	2	Whitefly, Thrips	Deep learning	TPest-RCNN	Validation (IoU = 0.7): mF1 0.944, mAP 0.952	NR	NR	CPU: Intel® Xeon® E5-2650 v4 (12 cores, 2.20 GHz), GPU: NVIDIA Tesla K80 (24 GB memory, 4992 CUDA cores)	NR	NR	labellmg	Public
[92] (2023)	Automatic pest detection and long-term counting	Cherry tomato	Yellow sticky paper	6	Tobacco whiteflies, Thrips, Winged aphids, Leaf miners, Fruit flies, Houseflies	NR	Yolov5 detector (focus/csp/spp, panet, ciou) + diou-nms	Precision 96%. Class F1: 0.99 (aphids, leaf miners, fruit flies, houseflies), 0.98 (whiteflies), 0.91 (thrips).	YOLOv5 = 0.83 s; Improved YOLOv5 = 7.53 s.	NR	CPU: Intel Core i7-7500U; GPU: NVIDIA GeForce 940MX, RAM: 8 Gb	Sony IMX226	Trap 25 × 20 cm: RGB 3280 × 2464, 1 image/day @16:00; train: 20 originals → 450 mosaics (600 × 600) with 200 copy-pasted; labels: 5800 instances (6 classes); test: 80 images (12,014 targets, all used)	NR	On request
[93] (2020)	Automatic detection, recognition, and counting	Tomato seedlings	Yellow sticky traps	4	Fly (Drosophilidae), gnat (Sciaridae), Thrips (Thripidae), Whitefly (Aleyrodidae)	Deep learning	Tiny-YOLOv3	Same greenhouse: mF1 = 0.92; ACC = 0.91, Different greenhouse: mF1 = 0.90; ACC = 0.90.	2.38 s/2.57 sNR		CPU: Intel Core i7-6700, GPU: NVIDIA GTX 1060	Raspberry Pi v2 camera	NR	Labellmg	On request

NR: Not reported.

3.4. Plant Growth Monitoring and Yield Prediction in SGHs

Plant growth monitoring and yield estimation in smart greenhouses increasingly leverage complementary data streams. Vision-based phenotyping provides interpretable proxies such as fruit presence, maturity state, and size-related measurements derived from detection or segmentation outputs. Time-series forecasting links historical production records with environmental variables to predict short-term yield trajectories. Combining these approaches enhances robustness, as image-derived proxies provide direct agronomic meaning while temporal models reduce variability caused by occlusion and intermittent sensing.

In vision-based monitoring, lightweight detection models have been integrated into IoT-enabled greenhouse pipelines for continuous tomato observation, demonstrating practical feasibility for in situ decision support [96]. Context-aware growth monitoring couples image recognition with environmental and temporal reasoning to stabilize interpretation under variable greenhouse conditions [97]. Dense segmentation has been used to convert organ-level pixel measurements into growth curves, with Gompertz fitting providing stable trend estimation for mini-tomato growth [98]. Similar machine learning-based monitoring approaches have been applied to other protected crops, including mushrooms, where stage-aware recognition supports operational harvest planning [99].

For yield and growth forecasting from sensor time series, deep temporal models predict short-horizon tomato yield from greenhouse environmental factors, with temporal convolution and recurrent architectures improving accuracy compared to single-sequence models [100]. Process-informed learning has been achieved by fusing a reduced biophysical crop model with CNN–RNN predictors, yielding higher predictive performance than either component alone [101]. Multi-step crop growth state prediction has been addressed using SVR Seq2Seq formulations, enhancing stability and generalization across prediction steps and crop datasets [102]. Table 5 summarizes the reviewed studies on plant growth monitoring and yield estimation in SGH systems; further details are presented in Table A4 provided in Appendix A.

Table 5. Summary of plant growth monitoring and yield estimation studies in SGHs.

Ref./Year	Crop	Task	Growth Stage	Sensors	Inputs	Interval	Dataset	Study Period	Model Family	Architecture	Performance	Dataset Availability
[96] (2025)	Tomato Normal/cherry	Fruit ripeness/object detection + classification	green, half-ripened, fully-ripened	DHT-11, LDR, SHT-10, Pi Camera	RGB images, Temperature, Humidity, Light intensity, Soil moisture	NR	804 images (3024 × 4032, 3120 × 4160 resized to 640 × 640)	NR	deep learning	YOLOv8	mAP: 52.8% (avg recall 0.478), Test accuracy: 52%, Per-class: b = cherry, l = normal b_fully_ripened: Pr = 0.444, R = 0.344, F1 = 0.388, Supprot = 93 b_half_ripened: Pr = 0.362, R = 0.313, F1 = 0.335, Supprot = 134 b_green: Pr = 0.625, R = 0.656, F1 = 0.640, Supprot = 369 l_fully_ripened: Pr = 0.465, R = 0.433, F1 = 0.448, Supprot = 289 l_half_ripened: Pr = 0.453, R = 0.419, F1 = 0.435, Supprot = 241 l_green: Pr = 0.489, R = 0.708, F1 = 0.578, Supprot = 638	On request
[97] (2024)	Tomato	Instance segmentation + maturity classification + Correcting missed fruit counts + harvest readiness	Fruit detection(maturity classes): Green, Half-ripened, Fully ripened Plant condition(growth stage): 0 = Before ripening, 1 = Ripe in 1–3 weeks, 2 = Ripe within 1 week, 3 = Fully ripened, 4 = No detection	Logicool C922n Pro webcam + CMUcam5 Pixy2	RGB images	NR	Training: 392 images	10/2021–06/2022	CNN + Bayesian network	YOLACT + DB-SCAN + Bayesian network	Fruit detection: Overall F1 = 0.84	Public
[98] (2022)	Mini-tomato (Milla)	growth monitoring	Flowering/fruit appearance→full ripening (rapid: day 5–25; slow: >25)	Raspberry Pi Zero W + 5MP camera + LED flash	RGB images (cropped 1024 × 1024), binary masks	15 min (00:00–18:00); modeling: 4 imgs/day (03:00, 05:30, 12:00, 16:30)	11,232 captured; MTIL: 385 labeled images + masks	18/07/2019–28/10/2019	DL semantic segmentation + nonlinear regression	TommyNET (modified symmetric U-Net w/multi-scale residual blocks) + Gompertz	Segm best: Precision = 0.99, Recall = 0.97, IoU = 0.95; Growth fit: $R^2 = 0.97$, sMAPE = 10.1%	Public
[99] (2023)	Mushroom	Growth-stage detection, Growth monitoring (size + harvest alert)	-Stage1/Stage2/Stage3 (Stage3 = ready to harvest) -Mask pixel area → growth rate + harvest time	Stage1 = first days/very small; Stage2 = formed, growing (not ready); Stage3 = cap edges flat/en/slightly up-rolled (ready)	RGB images (640pixels)	NR	1128 labeled imgs, 4271 instances (S1 = 1130, S2 = 1845, S3 = 1296); +231 photo monitoring (33 bags × 7 days); Detectron2 train sets: 453 mushrooms + 200 bags imgs	16/08/2022–24/08/2022	DL object detection + instance segmentation	YOLOv5s/YOLOv5l (default + hyperparameter evolution); Detectron2 (2 models: bag det + mushroom instance masks)	YOLOv5 (v5l + evolved HP + bs = 8): mAP@0.5 = 0.79353; mAP@0.5:0.95 = 0.54582, Precision = 77.89%, Recall = 75.44%, F1 = 76.65%, Per-class acc: Stage1 = 82%, Stage2 = 71%, Stage3 = 70%, Monitoring: avg growth period = 5.22 d; harvest earlier for 17 mushrooms (=14.04%); size threshold = 5.34%	On request
[100] (2021)	Tomato	Yield forecasting	NR	NR	recorded yield, CO ₂ , Temperature, Humidity deficit, RH, radiation	Daily; sliding window = 7 days, step = 1 day	GH1 (2018), GH2 (2017), GH2 (2018)	2017–2018	Deep learning (sequence)	LSTM + TCN + FC	RMSE (g/m ²): GH1: 10.45 ± 0.94; GH2: 6.76 ± 0.45; GH3: 7.40 ± 1.88	On request

Table 5. Cont.

Ref./Year	Crop	Task	Growth Stage	Sensors	Inputs	Interval	Dataset	Study Period	Model Family	Architecture	Performance	Dataset Availability
[101] (2023)	Tomato	Yield Prediction	Full season	NR	recorded yield, CO ₂ , temperature, humidity deficit, RH, radiation	NR	2-year UK grower	NR	Biophysical + Deep learning + Fusion ensemble	Reduced TOMGRO + CNN-RNN + fusion (linear/Bayes/NN/RFR/GBR)	RMSE = 17.69 ± 3.47 g/m ² , R ² = 0.9995 ± 0.0002, NSE = 0.9989 ± 0.0004, PBIAS = 0.1791 ± 0.6837	On request
[102] (2025)	cabbage, lettuce, spinach	crop growth multi-step prediction(Plant height, plant width)	NR	NR	air humidity, air temperature, light intensity, soil moisture content, soil temperature, soil electrical conductivity, pH value, Nitrogen content, Phosphorus content, Potassium content	Env: 1 min; Growth-state: 3×/day (09:00, 13:00, 17:00)	cabbage: 03/11/2023–25/11/2023, lettuce: 19/11/2023–13/12/2023, spinach: 08/12/2023–31/12/2023)	11/2023–01/2024	Hybrid ML + DL	SVR_Seq2Seq: SVR (RBF kernel) + Seq2Seq (Bi-LSTM encoder + LSTM decoder) + linear fusion + FC (output layer)	Height (cm): Next day (k = 3): MAE = 0.1239, RMSE = 0.1357, MAPE = 0.0123 2 days ahead (k = 6): MAE = 0.1500, RMSE = 0.1539, MAPE = 0.0147 3 days ahead (k = 9): MAE = 0.1881, RMSE = 0.1973, MAPE = 0.0185 4 days ahead (k = 12): MAE = 0.2242, RMSE = 0.2503, MAPE = 0.0226 Width (cm): Next day (k = 3): MAE = 0.3139, RMSE = 0.3281, MAPE = 0.0209 2 days ahead (k = 6): MAE = 0.3794, RMSE = 0.4063, MAPE = 0.0255 3 days ahead (k = 9): MAE = 0.4535, RMSE = 0.5237, MAPE = 0.0315 4 days ahead (k = 12): MAE = 0.4918, RMSE = 0.5347, MAPE = 0.0347	On request
[103] (2021)	sweet pepper	Fruit detection, fruit growth prediction	IM1: 35 days before harvest IM2: 25–35 days before harvest IM3: 15–25 days before harvest IM4: 15 days before harvest B (Breaker): 7 days before harvest M (Mature): harvest stage	RGB images, env time-series	RGB images, env + growth	NR	7682 images (resized 1024 × 1024) 35,958 fruits labeled	26/02/2020–07/07/2020, 06/04/2020–24/06/2020	DL detection (CNN) + DL tabular classifier (MLP) + ensemble	Yolov5, MLP	Ensemble (6 stages): mean F1 = 0.77, IoU = 0.86; CNN-only means F1 = 0.56; MLP-only means F1 = 0.49; CNN (3 stages in ensemble): F1 = 0.91, IoU = 0.86	NR
NR: Not reported.												

3.5. Robotic Harvesting in SGHs

Robotic harvesting in smart greenhouse systems has emerged as a key technology for automating labor-intensive crop management operations, including fruit picking, pruning, and maturity monitoring. These systems aim to reduce labor dependence, address workforce shortages, and improve operational efficiency in controlled-environment agriculture. Most robotic harvesting platforms integrate four core modules: autonomous navigation, environmental mapping, automatic control, and machine vision-based perception. Autonomous navigation enables robots to determine their precise location within greenhouse environments, plan collision-free paths, and adapt motion to dynamic conditions. Positioning and mapping are typically achieved through multi-sensor data fusion, combining inputs from vision sensors, LiDAR, radar, and infrared modules. These data are processed using control and localization algorithms to generate accurate spatial maps and support real-time motion adjustment. Machine vision represents the central component of robotic harvesting systems, supporting perception tasks such as fruit detection, ripeness assessment, key-point localization, and three-dimensional pose estimation. Early systems relied on conventional image processing methods, whereas recent approaches increasingly employ deep learning-based detectors, including YOLOv4, YOLOv5, Mask R-CNN, and YOLACT++, often enhanced through attention mechanisms, multi-scale feature fusion, and stereo vision for depth estimation. For instance, strawberry harvesting robots using DHN-YOLO models achieved an mAP50 of 91.8% and processing speeds of 71.6 FPS on images captured using mobile devices [104]. In tomato harvesting applications, YOLOv5 integrated with stereo depth cameras reached detection accuracies of up to 94.1% while enabling real-time three-dimensional fruit localization [105]. Similarly, multi-joint robotic manipulators equipped with RGB-D sensors have demonstrated successful detection of flowers and sweet peppers, with harvesting success rates ranging from 57% to 93.3%, depending on crop type and occlusion levels [106,107]. Table 6 summarizes key robotic harvesting systems in smart greenhouses, including their sensing technologies, perception tasks, and reported performance; further details are presented in Table A5 provided in Appendix A.

Table 6. Summary of vision-based robotic harvesting systems in SGHs.

Ref./Year	Crop	Robot Platform	Architecture	Performance	Sensors	Perception Task	Success Rate	Dataset	Dataset Availability
[104] (2025)	Strawberry	NR	DHN-YOLO (YOLO11n-pose + CDC(CGA+DCNv3) + C3H(HetConv) + New-Neck	Fruit detection: P = 87.3%, R = 88.0%, mAP50 = 91.8% mAP50:95 = 78.6%; KeyPoint detection: P = 83.0%, R = 87.5%, mAP50 = 89.7%, mAP50:95 = 83%	Smartphone rear camera	Maturity-stage detection (unripe/ripe/fully ripe) + key picking point detection	NR	2066 images (1024 × 768 augmented to 5018 images)	On request
[105] (2025)	Tomato	Vision-based detection + 3D localization module for a tomato picking robot	Detection/YOLOv5 + stereo depth/3D position/SGBM; training uses SGD + Mosaic + warmup + cosine annealing	NR	Lena CV USB3.0 binocular camera (stereo)	Tomato detection in 3 classes: unoccluded/leaf-occluded/branch-occluded + 3D localization (depth) using stereo matching	Detection: 94.1% accuracy (YOLOv5; 95/102 correct) and mAP@0.5 = 0.947 (class mAPs: 0.941/0.953/0.946)	640 greenhouse images collected (0.7 m camera distance) augmented to 7788 images	On request
[106] (2024)	Sweet pepper	Universal Robot UR3 (6-DOF) controlled with MoveIt + RRT-Connect, ROS Noetic modules	Real-time semantic segmentation NN (MobileNetV2 bottleneck blocks + pyramid pooling; depth explored); 3D recon uses ORB-SLAM3 + ICP registration	NR	Intel RealSense L515 RGB-D camera	Semantic segmentation of plant parts (stem/leaf/petiole/fruit/others) → 3D semantic point cloud → detect pruning position + pruning direction (end-effector pose)	57% success over 30 attempts (43% failures with reasons reported)	1000 sweet-pepper images	On request
[107] (2021)	Cherry tomato	Mobile robot platform + robotic arm; eye-in-hand RGB-D camera. Workflow: detect bunch at 0.5–0.7 m, then arm moves camera to 0.3 m close-up for peduncle segmentation and pose estimation	Stage-1: YOLOv4-Tiny for tomato-bunch detection (coarse), Stage-2: YOLACT++ instance segmentation (R-101-FPN chosen) to segment bunch & peduncle masks, then least-squares curve fit + 3 keypoints + geometric model for pose	Vision results: YOLOv4-Tiny Precision 92.7%, Recall 96.2%, F1 94.4%, speed 9.1 ms/img, YOLACT++ (R-101-FPN): mAP 73.1, mAP50 96.9, mAP75 87.3, 9.2 FPS (0.109 s/frame)	Intel RealSense D435i & D415 (RGB-D)	Peduncle cutting-point localization + peduncle pose estimation (yaw/pitch) using 2-stage detection/segmentation and geometric fitting	Peduncle keypoints found in 112/120 close-up tests (93.3%), Pose error (30 tests): yaw 4.98°, pitch 4.75° avg	1528 images total = 828 long-distance (0.5–0.7 m) + 700 close-up (0.25–0.35 m)	NR
[108] (2024)	Marigold, Snapdragon, Pansy/horned pansy	Straddle cart over greenhouse benches (120 cm wide, 175 cm high) with Jetson TX2 for edge processing	Detection: YOLOv5l fine-tuned (FLOLO) + Segmentation: SAM (ViT-Huge); Pose: PCA on segmented point cloud; Plucking point: linear regression	NR	Stereolabs ZED 2/ZED 2i RGB-D stereo cameras	Ripe flower detection (2D), segmentation, 3D localization, pose estimation, plucking-point estimation	Detection (FLOLO test): mAP@0.5 = 0.68, P = 0.67, R = 0.68	FloraDet: 134 images/2443 flowers	On request
[109] (2024)	Strawberry	Wire-driven multi-joint arm with spherical joints (3D printed), mounted on lifting module; module moves along aluminum extrusion track on hydroponic shelf; deployed on Jetson Nano; Arduino-based actuation mentioned	YOLOv4 detector (CSPDarknet53 + SPPNet + PANNet)	Detection: precision 0.891, recall 0.880, AP@0.5 = 0.891, AP@0.5:0.95 = 0.718 (avg over test periods), 90% detection accuracy	2× RGB cameras (Logitech Brio Ultra 4K HD)	NR	Up to 82% fruit picking success rate (FPR); as low as 32% early cloudy morning	2000 RGB images (416 × 416 pixels)	NR

Table 6. Cont.

Ref./Year	Crop	Robot Platform	Architecture	Performance	Sensors	Perception Task	Success Rate	Dataset	Dataset Availability
[110] (2024)	Tomato	Six-degree-of-freedom & Tracked mobile chassis, 5-degree-of-freedom robotic arm, lifting mechanism, collection basket, and laser radar	YOLOv5, HSV + fusion for mature-red extraction; night-time denoising uses Gaussian + guided filtering	NR	ZED 2i stereo camera + laser radar RS-Helios-16P	RGB, depth map, 3D point cloud (stereo)	Day: 87.78%; Night: 87.55%	Self-collected images: 8 images per tomato cluster from multiple angles (10°, 45°, 90°, 135°, 170°, plus −45°, 0°, 45°); scenes include single/multiple tomatoes with/without occlusion;	NR
[111] (2024)	Tomato	Rail cart system with vertical/horizontal mast + 2 fixed cameras (30 cm apart): Perpendicular camera (PC) + Angled camera (AC); an extra eye-in-hand camera dataset collected from a robot arm	Mask R-CNN (Detectron2)	NR	Intel RealSense D435 cameras	Instance segmentation/detection of tomatoes + stem parts	NR	Pose-PC: 100 labeled images (from 100 image pairs), 1080 × 1920, perpendicular view. Pose-AC: 100 labeled images, 1080 × 1920, angled view (25°). Illumination: 175 images, 2048 × 2448, collected in 2017 with poor illumination/overcast. Task: 184 labeled images, RealSense on robot arm, angled 54°, wider perspective	On request

NR: Not reported.

3.6. Evaluation Metrics for AI Applications in SGHs

Classification papers typically use accuracy together with precision, recall, and F1, and may add AUC or TSS when class imbalance is a concern. Detection and instance-segmentation studies mainly rely on IoU and AP-based scores, most commonly mAP with fixed or COCO-style IoU thresholds. Regression and forecasting are usually assessed with MAE and RMSE, sometimes complemented by percentage errors and fit indicators such as R^2 . For edge and real-time deployments, latency, throughput, model size, and compute cost are increasingly reported to support accuracy efficiency comparisons. Table 7 consolidates the most commonly used metrics in the reviewed SGH literature.

Table 7. Common evaluation metrics reported in AI-enabled SGH studies, grouped by task type.

Task Type	Metric	Meaning, Equation, and Typical Interpretation in SGH Studies
A. Classification/recognition		
Classification	Accuracy (Acc)	Measures the overall proportion of correctly classified samples among all predictions. It is easy to interpret and is commonly used in crop disease recognition, pest identification, and growth-stage classification. However, it can be misleading when the dataset is imbalanced and one class dominates the others. $Acc = \frac{TP+TN}{TP+TN+FP+FN}$.
Classification	Precision (P)/Recall (R)/Specificity (Sp)	These metrics provide class-sensitive evaluation. Precision shows how many predicted positives are truly positive and is useful when false alarms should be minimized. Recall indicates how many true positives are successfully detected and is important when missing diseased or stressed plants is costly. Specificity reflects how well the model identifies true negatives, such as healthy plants or non-target classes. $P = \frac{TP}{TP+FP}$, $R = \frac{TP}{TP+FN}$, $Sp = \frac{TN}{TN+FP}$.
Classification	F1-score (F1)	Harmonic mean of precision and recall. It is especially informative in SGH studies with imbalanced datasets because it balances missed detections and false positives rather than favoring only one of them. Commonly reported in disease, nutrient-deficiency, and stress classification tasks. $F1 = \frac{2PR}{P+R}$.
Classification (multi-class)	Macro-averaged P/F1	Computes the metric independently for each class and then averages across all classes. This is useful in multi-class SGH problems because each category receives equal importance, even when some classes have many fewer samples than others. $mP = \frac{1}{C} \sum_{c=1}^C P_c$, $mF1 = \frac{1}{C} \sum_{c=1}^C F1_c$.
Classification	AUC-ROC	Evaluates the model's ability to separate positive and negative classes across all decision thresholds. A higher AUC indicates better ranking performance and threshold-independent discrimination, which is useful when comparing competing classifiers. $AUC = \int_0^1 TPR(FPR) dFPR$.
Classification (imbalanced)	True Skill Statistic (TSS)	Measures classification skill beyond random chance by combining sensitivity and specificity. It is particularly useful for imbalanced SGH datasets where rare but important classes, such as early disease symptoms, must still be detected reliably. $TSS = Sen + Sp - 1$.
B. Object detection/instance segmentation		
Detection	IoU	Quantifies the spatial overlap between the predicted bounding box and the ground-truth box. Higher IoU means better localization quality. In SGH studies, it is widely used for fruit, leaf, flower, and pest detection. $IoU = \frac{ B \cap \hat{B} }{ B \cup \hat{B} }$.
Detection	AP/mAP	AP summarizes the precision–recall trade-off for one class, while mAP averages AP across all classes. These are standard metrics for detection tasks because they assess both whether objects are detected and how reliably detections are ranked. In SGH applications, they are frequently used in harvesting and monitoring systems. $AP = \int_0^1 p(r) dr$, $mAP = \frac{1}{C} \sum_{c=1}^C AP_c$.
Detection	mAP@0.5 and mAP@0.5:0.95	mAP@0.5 considers a detection correct when IoU is at least 0.5, while mAP@0.5:0.95 averages performance over stricter IoU thresholds. The latter gives a more demanding evaluation and better reflects localization robustness in practical greenhouse scenes.
Detection	Recall@t	Measures the proportion of true objects detected under a given matching rule, often with $IoU \geq t$. This metric is useful when the priority is to avoid missing relevant targets such as fruits, lesions, or insects. $R@t = \frac{TP_t}{TP_t+FN_t}$.

Table 7. Cont.

Task Type	Metric	Meaning, Equation, and Typical Interpretation in SGH Studies
C. Segmentation/severity estimation		
Segmentation	Mean IoU/Mean Jaccard	Evaluates how well the predicted mask overlaps with the reference annotated region. It is widely used in semantic or instance segmentation of leaves, fruits, canopy regions, and diseased areas. Higher values indicate more accurate delineation of object boundaries. $mIoU = \frac{1}{N} \sum_{i=1}^N \frac{ S_i \cap \hat{S}_i }{ S_i \cup \hat{S}_i }$
Severity estimation	Mean Absolute Error of severity	Measures the average absolute difference between predicted and true severity scores. It is appropriate when disease level, stress intensity, or damage severity is represented on a continuous or ordinal numerical scale. Lower MAE indicates more accurate estimation. $MAE = \frac{1}{N} \sum_{i=1}^N s_i - \hat{s}_i $
D. Regression/forecasting		
Regression	MAE/RMSE	MAE reports the average magnitude of prediction error, while RMSE penalizes large deviations more strongly. These metrics are commonly used in SGH studies for temperature, humidity, yield, irrigation demand, and nutrient prediction. MAE is easier to interpret, whereas RMSE is more sensitive to large mistakes. $MAE = \frac{1}{N} \sum_{i=1}^N y_i - \hat{y}_i $, $RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$
Regression	MAPE or relative error in %	Expresses prediction error relative to the true value, making it useful for comparing performance across variables with different scales. It is often reported for greenhouse climate and production forecasting, although it should be interpreted carefully when true values are close to zero. $MAPE = \frac{100}{N} \sum_{i=1}^N \left \frac{y_i - \hat{y}_i}{y_i} \right $.
Regression	R^2 or Pearson R	These metrics describe goodness of fit and the strength of association between predictions and observations. R^2 indicates the proportion of variance explained by the model, while Pearson R reflects linear correlation. They are usually reported together with MAE or RMSE rather than alone. $R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$, $R = \frac{\sum_i (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_i (y_i - \bar{y})^2} \sqrt{\sum_i (\hat{y}_i - \bar{\hat{y}})^2}}$.
E. Deployment and efficiency reporting		
Efficiency	Latency and throughput	Latency measures the time required to process one sample or frame, while throughput describes the number of samples processed per unit time, often reported as FPS. These metrics are essential for real-time SGH monitoring, robotic harvesting, and edge-based inference.
Efficiency	Model size and parameters	Reflect the memory footprint and the number of trainable parameters of the model. These are important when comparing lightweight and high-capacity models for embedded greenhouse devices with limited storage and memory resources.
Efficiency	FLOPs or MACs	Estimate the computational cost of one inference pass. Lower values generally indicate a more efficient model, which is important for low-power deployment and real-time greenhouse decision systems. FLOPs are often reported as GFLOPs, while MACs represent multiply-accumulate operations.

Notes: TP , TN , FP , and FN denote true/false positives/negatives. y_i and $\hat{y}_i$ are ground-truth and predicted values; s_i and $\hat{s}_i$ are true and predicted severity scores; N is the number of samples; C is the number of classes; B and $\hat{B}$ are ground-truth and predicted boxes; S and $\hat{S}$ are ground-truth and predicted masks.

3.7. Data Types and Acquisition Tools in SGHs

Across the reviewed SGH studies, data collection is dominated by three sources, as summarized in Table 8. First, proximal imaging, mainly RGB, is the most widely used due to its low cost and its direct relevance for monitoring and phenotyping. Second, proximal spectral sensing, including multispectral, hyperspectral, and thermal sensing, is adopted when the aim is to capture early physiological responses that may appear before visible symptoms. Third, distributed IoT/WSN sensing enables continuous monitoring of the greenhouse microclimate.

Accordingly, the collected data in SGH studies largely fall into two families. The first family is spatial observations, including RGB, RGB-D, thermal, multispectral, and hyperspectral imagery, commonly used for detection, segmentation, and phenotyping. The second family is sensor time series, including temperature, humidity, CO_2 , and light or radiation, and when available, root-zone and fertigation variables such as moisture, pH, and EC, which are mainly used for forecasting and optimization. Table 8 provides a consolidated view linking acquisition sources to the resulting data types, while the public datasets used in the reviewed papers and their access links are reported in Table 9.

Table 8. SGHs acquisition platforms and typical input modalities reported in the reviewed SGH literature.

Acquisition Platform	Data Type	Typical SGHs Use
Ground-based imaging, fixed or handheld	RGB images or video	Phenotyping and growth tracking, symptom recognition, fruit and leaf detection, visual scouting.
	Trap imagery, sticky or light traps	Pest detection, classification, and abundance estimation for IPM decision support.
	RGB-D or stereo depth	Geometry-aware perception, 3D localization and sizing, occlusion-robust detection, harvesting and manipulation.
	Thermal or IR imagery	Canopy temperature monitoring for water-stress and transpiration-related anomalies.
Proximal spectral sensing	Multispectral imaging	Stress and disease characterization using selected VIS–NIR bands with moderate acquisition complexity.
	Hyperspectral imaging	Fine-grained physiological signatures and early stress detection with higher calibration burden.
	Reflectance spectra, spectrometers	Point measurements for spectral indices and biochemical or stress proxies without full imaging.
In situ environmental sensing, IoT or WSN	Air microclimate time series, temperature, humidity, CO ₂ , VPD	Microclimate monitoring, forecasting, anomaly detection, and climate-control modeling.
	Radiation and light, PAR, PPFD, solar radiation	Light-driven growth modeling and energy-aware control.
	External or adjacent weather time series.	Exogenous inputs for indoor climate prediction when combined with actuation data.
Robotic platforms for scouting/harvesting	On-board perception, RGB, RGB-D, LiDAR, IR; localization signals, IMU, GPS where applicable	Mobile data collection, navigation support, scouting, task execution in crop rows, pose estimation, trajectory repeatability, and autonomous operation.

Table 9. Summary of publicly available datasets.

Ref.	Dataset Name	Crop	No. of Classes	Target	Data	Dataset URL	Task
[32]	GreenhouseData	Tomato, Bell pepper	1	Internal Air temperature	85,989 samples	https://github.com/fabiangarciauaaz/GreenhouseData (accessed on 15 February 2026)	Climate prediction
[49]	Climate and Crop variables of tomato greenhouse simulation	NR	1	Internal air temperature	NR	https://github.com/fabiangarciauaaz/GreenhouseData (accessed on 15 February 2026)	climate prediction
[71]	Strawberry powdery mildew	Strawberry	3	powdery mildew regions, infected leaf regions, noninfected leaves	1023 images (1040 × 780)	https://github.com/liyang166/DAC-YOLOv4 (accessed on 15 February 2026)	diseases detection
[75]	Strawberry Disease Detection Dataset	strawberry	7	Angular Leafspot, Anthracnose Fruit Rot, Blossom Blight, Gray Mold, Leaf Spot, Powdery Mildew Fruit, and Powdery Mildew Leaf	2500 RGB images (419 × 419 resolution)	www.kaggle.com/usmanafzaal/strawberry-disease-detection-dataset (accessed on 15 February 2026)	diseases detection
[78]	NR	Tomato, Cucumber, Eggplant	20	Tomato (10): Healthy, Early blight, Late blight, Gray mold, Leaf mildew, Leaf spot, Ulcer disease, Anthracnose, Leaf curl, Viral disease Cucumber (6): Healthy, Powdery mildew, Downy mildew, Brown spot, Anthracnose, Viral disease Eggplant (4): Healthy, Verticillium wilt, Brown spot, Viral disease	Partially public/full on request	https://github.com/tyuiuioio/plant-disease-detection-in-real-field (accessed on 15 February 2026)	diseases detection
[79]	NR	Barley	NR	NR	Partially public/full on request	https://openxlab.org.cn/datasets/wyz123/cropper (accessed on 15 February 2026)	diseases detection
[82]	vegetable disease	Pepper	8	healthy, anthracnose, phytophthora blight, viral disease, leaf spot disease, root rot, blossom-end rot, mites, and thrips	8046 images (640 × 640)	https://data.mendeley.com/datasets/tg3z7xxkdb/1 (accessed on 15 February 2026)	disease and pest detection
[83]	vegetable disease	Cucumber	5	Healthy, Anthracnose, Bacterial Wilt, Pythium Fruit Rot, Downy Mildew	Partially public/full on request (4740 images total (Healthy 1630; Anthracnose 880; Bacterial Wilt 790; Pythium Fruit Rot 750; Downy Mildew 690))	https://data.mendeley.com/datasets/tg3z7xxkdb/1 (accessed on 15 February 2026)	disease lesion object detection
[85]	vegetable disease	Tomato	5	ate blight, gray leaf spot, brown rot, leaf mold, healthy	2000 images (captured at 3648 × 2056) augmented to 9600 images	https://github.com/tyuiuioio/plant-disease-detection-in-real-field (accessed on 15 February 2026)	Tomato disease object detection
[90]	PST Pest Sticky Traps	NR	1	whitefly (2 species: Bemisia tabaci, Trialeurodes vaporariorum)	28 RGB images (4288 × 2848 resolution), 17,005 annotated whiteflies	https://zenodo.org/records/7801239 (accessed on 15 February 2026)	Insect counting/population density estimation on sticky traps

NR: Not reported.

4. Spectral Sensing in SGHs: HSI and MSI with ML and DL

Advances in sensor technologies have enabled crop monitoring across a broad range of electromagnetic wavelengths beyond the visible spectrum [112]. Spectral imaging captures reflectance information in non-visible bands that contains more discriminative physiological signals than RGB imaging, particularly for detecting early biochemical and structural plant changes induced by biotic stress [113].

Multispectral imaging and hyperspectral imaging provide complementary characteristics. Hyperspectral imaging offers dense narrow-band spectral information that is highly sensitive to subtle pre-symptomatic plant responses, whereas multispectral imaging provides fewer spectral bands but higher spatial resolution, which is advantageous once symptoms become spatially visible [114,115].

The integration of spectral imaging with machine learning and deep learning has significantly improved plant monitoring capabilities. In disease detection, hyperspectral imaging is effective for identifying early physiological responses prior to visible symptoms, while multispectral imaging is more suitable for detecting spatially expressed lesions. This complementarity was demonstrated in Dutch elm disease detection, where hyperspectral data enabled earlier discrimination, whereas multispectral data improved identification of symptomatic leaf regions [116]. However, model performance can be affected by spectral variability across genotypes. In Potato Virus Y detection, artificial neural network classifiers achieved high accuracy when trained on a single variety but showed substantial performance decline when applied to multiple genotypes [117]. In pest detection applications, multispectral imaging combined with deep learning commonly follows a spectral-spatial detection framework. Incorporating non-visible channels such as near-infrared bands improves lesion detection performance compared with RGB-only models, as demonstrated by multichannel Faster R-CNN architectures that integrate RGB and near-infrared features [118].

For weed monitoring and herbicide resistance detection, studies frequently employ wavelength selection and vegetation index development. Spectral reflectance data combined with feature selection methods enables identification of discriminative wavelengths associated with herbicide-induced biochemical changes. Optimized spectral weed indices derived from selected wavelengths have demonstrated high classification accuracy in detecting glyphosate-resistant weed species under greenhouse conditions [119,120].

A similar approach has been applied to early pest infestation detection. Hyperspectral imaging combined with preprocessing and wavelength selection has enabled accurate classification of infestation severity levels in cotton leaves, with near-infrared wavelengths showing strong sensitivity to mite-induced plant damage [121,122].

Table 10 summarizes the reviewed HSI and MSI studies, and Table 11 reports the key effective wavelengths and vegetation indices.

Table 10. Overview of hyperspectral and multispectral imaging studies applied in smart greenhouse environments.

Study/Year	Wavelength Range (nm)	Type of Spectral Data	Sensor Type	Target Crops	Application
[116] (2023)	490–900	Hyper-spectral and multi-spectral data	High-precision leaf imagers (LeafSpec)	American elm (<i>Ulmus americana</i>)	Detection of Dutch elm disease (DED) and resistance screening
[117] (2023)	350–2500	hyperspectral (Electromagnetic (EM) reflectance)	FieldSpec 4 spectroradiometer (Analytical Spectral Devices, a PANalytical Company, Longmont, CO, USA)	Potato (<i>Solanum tuberosum</i>)	Detection of PVY-infected plants
[118] (2023)	RGB (460, 540, and 630), NIR (850 and 980)	RGB and multispectral imaging	MUSES9-MS-PL multispectral camera (Spectricon, Chania, Greece)	tomato (<i>Solanum lycopersicum</i>) plants	the early and accurate detection of Tuta absoluta and Leveillula taurica
[119] (2023)	460–980	multispectral data	MAPIR Survey 3 camera	Cabbage (<i>Brassica oleracea</i> var. <i>capitata</i>)	Estimation of biometric, physiological, and nutritional parameters in cabbage seedlings
[120] (2020)	450–920	multispectral data	NR	ragweed (<i>Ambrosia artemisiifolia</i> L.), and waterhemp (<i>Amaranthus rudis</i>).	The early detection of glyphosate-resistant weeds
[122] (2024)	400–1000	hyperspectral data	a benchtop hyperspectral sensor (PIKA L, Resonon Inc., Bozeman, MT, USA)	cotton plants	Early detection and classification of Tetranychus urticae infestation levels.
[123] (2023)	400–1000	hyperspectral imaging data	HSC-2 SENOP camera	Cucumber	Predicting transpiration under CO ₂ enrichment
[124] (2025)	400–1000	non-imaging hyperspectral reflectance	Stellar Rad + Color Spectroradiometer (StellarNet Inc., Tampa, FL, USA)	Chenopodium album (common lambsquarters)	Detection and quantification of herbicide (glyphosate) injury
[125] (2021)	400–1000	hyperspectral imaging	SPECIM IQ Hyperspectral Camera (SPECIM, Spectral Imaging Ltd., Oulu, Finland)	Wild Rocket (<i>Diplotaxis tenuifolia</i>)	Non-destructive detection of powdery mildew disease
[126] (2022)	400–1000	hyperspectral data	SPECIM IQ hyperspectral camera (SPECIM, Spectral Imaging Ltd., Oulu, Finland)	Wild Rocket (<i>Diplotaxis tenuifolia</i>)	detecting biotic and abiotic stresses (Fusarium, Rhizoctonia, water deficit, and salinity) in leaves.
[127] (2020)	370–1030	hyperspectral data	MSV-500 hyperspectral cameras (Middleton Spectral Vision, Middleton, WI, USA)	Maize (<i>Zea mays</i> L.)	Relative Water Content estimation under drought stress
[128] (2021)	350–2500	hyperspectral reflectance spectra	FieldSpec 4 Hi-Res spectroradiometer (Analytical Spectral Devices, Boulder, CO, USA)	Rice (three varieties: Wuyungeng 7, Nipponbare, Nangeng 44)	Early detection of rice leaf blast disease (asymptomatic to mild stages)
[129] (2021)	200–1100 (used: 240–900)	hyperspectral	Jaz Spectrometer System (Ocean Optics, Dunedin, FL, USA).	Peanut (<i>Arachis hypogaea</i> L.)	Disease detection (stem rot by <i>Athelia rolfsii</i>)
[130] (2020)	350–2500	hyperspectral imaging	HR-1024i spectroradiometer (Spectra Vista Corporation, Poughkeepsie, NY, USA)	Snap Bean (<i>Phaseolus vulgaris</i>)	Yield prediction under greenhouse conditions

Table 11. Key effective wavelengths and vegetation indices reported in spectral imaging studies.

Study	Effective Wavelengths (nm)	Vegetation Indices
[119]	550 (Green), 660 (Red), 850 (NIR)	SR to estimate nutritional variables, NDVI for biometric parameters like leaf area and number of leaves, GNDVI assess plant mass
[123]	NVDI (RED: 680, NIR: 800), PRI (531, 570), WBI (900, 970)	NVDI: Indicates chlorophyll content and photosynthetic activity, PRI: Reflects light-use efficiency and early stress detection, WBI: Measures canopy water content and hydration status
[124]	432.5, 532.5, 587.5, 677.5, 682.5	Custom-developed VIs using combinations of these wavelengths (e.g., VI_{BF} and VI_{min} for glyphosate injury prediction
[126]	B5 (408.85 nm), B3 (403.09 nm), B6 (411.74 nm), B18 (446.45 nm)	Custom-developed VIs for detecting powdery mildew, based on the selected wavelengths
[127]	492–504 nm (VIS), 540–568 nm (VIS), and 712–720 nm (VIS), and 855 nm, 900–908 nm, and 970 nm for NIR region.	No VIs were mentioned in the study; instead, it focused on identifying specific spectral bands that were most effective for classifying plant stress and disease types.
[129]	501, 505, 686, 690, 694, 763, 830, 884	Not specified

5. Multimodal Data Fusion in SGHs

Multimodal data fusion has emerged as a promising way for SGH applications, as the state of crops and the greenhouse itself is naturally multimodal in nature. A single modality may not be able to completely represent the context of the crops and the greenhouse. The RGB images may provide some visual cues for symptoms and morphology, but it is easily affected by changes in illumination, occlusion, and backgrounds. Spectral images may detect changes in crops before symptoms appear, which is useful, although it may be costly and time-consuming. The environmental sensors continuously monitor temperature, humidity, CO₂, and soil status, etc., but cannot directly reflect the appearance of crops and abnormalities in organs. Therefore, data fusion in multiple modalities may help improve the robustness and relevance of decisions by effectively making use of multiple information sources. In SGHs, multimodal fusion is particularly interesting since there is a strong coupling between plant environment interactions. For example, disease occurrence, water stress, growth rate, and pest occurrences are generally not caused by vision or environmental sensing alone. Instead, they are often a function of coupled effects between plant physiology, microclimate, temporal history, and spatial information. This makes SGHs an interesting setting for multimodal AI, where vision, spectral, IoT, and temporal information can be combined within a unified framework. From a methodological point of view, multimodal fusion in SGHs can be classified into three types: data-level fusion, feature-level fusion, and decision-level fusion. At the data level, multiple sources of data are integrated before feature extraction. This is a popular choice in spectral research, where VIS/NIR and NIR signals are often combined into a unified space. The advantage of this method lies in its ability to preserve low-level information, which might allow the model to directly capture cross-modal correlations. Nevertheless, this method requires careful calibration, normalization, and synchronization, as noise or scale issues in one modality might affect the quality of the integrated data. Feature-level fusion is more commonly used, as it allows the separate processing of the different modality inputs by a specific encoder before their fusion. This type of fusion is most appropriate when the different modalities significantly differ from one another, as in the case of RGB images, environmental

data, and temporal variables. Decision-level fusion involves integrating predictions from different unimodal or partially fused models. This approach is advantageous when there are differences in rates of information capture, when a single modality is absent, or when a module-based approach is necessary. Although it is less expressive than learning a joint representation, it is also potentially more robust. In practical greenhouse deployments, this advantage is important because missing values, intermittent connectivity, and calibration drift are common. A second important aspect is temporal fusion. The information captured by a sensor in a greenhouse is often temporal, with a signal's usefulness depending on its history rather than its instantaneous value. For example, environmental conditions before a disease or pest outbreak are often critical. Temporally informed fusion combines multimodal information with sequence information. This is achieved through a variety of techniques, including recurrent neural networks, temporal convolutional networks, and attention-based approaches. These benefits have been demonstrated across several greenhouse tasks.

In disease detection, multimodal approaches integrating RGB imagery with spatio-temporal and environmental metadata consistently outperform single-modality models, providing higher accuracy under variable greenhouse conditions [131]. Spectral disease diagnosis also benefits from cross-sensor fusion, merging visible and near-infrared bands either at the raw-data or feature level, achieving high classification accuracy when regions of interest are properly aligned [132]. Similarly, in stress assessment and phenotyping illustrate additional benefits. Combining RGB and thermal data with growth indicators increases robustness to environmental variability [133]. Multi-sensor phenotyping that incorporates calibrated three-dimensional representations, such as camera-LiDAR alignment, enables more precise structural measurements [134]. For continuous monitoring, early fusion with ensemble models balances accuracy and computational efficiency [135]. Temporally aware fusion pipelines using one-dimensional convolutional feature extraction followed by sequence modeling reduce forecasting error for pest early-warning applications [136]. Overall, multimodal fusion is a key enabler of operationally reliable AI in smart greenhouses. By integrating complementary modalities, models achieve greater resilience to environmental variability, improved early detection, and more accurate monitoring, supporting scalable and data-driven greenhouse management.

As summarized in Table 12, multimodal fusion approaches in smart greenhouses vary in sensor types, data modalities, and learning methods.

Table 12. Summary of multimodal fusion approaches in smart greenhouses.

Ref./Year	Crop/Task	Modalities	Feature Extractors	Fusion Level	Fusion Method	Fusion Network	Alignment	Performance/Fusion Gain	Hardware	Inference Time/Model Size	Dataset/Availability
[131] (2024)	tomato, cucumber, bitter melon/disease detection	RGB images, multi-source metadata (env, time, space)	visual: CNN backbone + Swin Transformer, Text: BERT encoder	Feature-level fusion	Space-Time fusion attention(STFA) Multilayer Encoder–Decoder Feature fFusion (MEDFFN)	MIFV (STFAN + MEDFFN)	Space-time attention-based feature alignment	mAP = 92.38%, (+3.43% vs. YOLOv7-tiny, +3.02% vs. YOLOv8n)	GPU: NVIDIA GeForce 3060 Ti (32 GB)	43.6 FPS/39.07 M	Partly public/full on request
[132] (2024)	tomatoes/early diagnosis of <i>Cladosporium fulvum</i>	VIS/NIR HSI + NIR HSI	PCA, VPCA, IRIV	low-level data fusion, medium-level data fusion	Low-level data fusion: merge VIS/NIR + NIR spectra into a new dataset. Medium-level data fusion: PCA/VCPA/IRIV feature extraction on each block, then feature fusion	PCA-RBF	430–900 nm + 950–1650 nm; black-and-white correction; ROI (SpectralView)	low-level fusion): Acc 100% (cal)/99.3% (pred); Macro-F1 = 1; G-mean = 1; 14 feature wavelengths	NR	NR	1374 samples (687 VIS/NIR + 687 NIR); subset 605 (55/class) for calibration/prediction/upon request
[133] (2025)	Sweet potato/classify water stress levels	RGB imagery, TRI, growth indicators (stem length, NDVI, chlorophyll fluorescence, SPAD), CWSI, leaf temperature, soil moisture content, T_a , RH	CNN, Global Average Pooling, ViT	Feature-level (RGB + TRI) + Decision-level (CNN+ViT + KNN)	Averaging of prediction results from CNN+ViT and KNN models	CNN-based ViT model (CNN+ViT) + KNN model	NR	CNN+ViT: 0.92 (accuracy); 0.91 (5-fold cross-validation average)	GPU: Nvidia GeForce GTX 1060	-/122 MB (fp32)	904 images (452 TRI [grayscale], 452 RGB; resized to 128 × 128) + 300 integrated data samples/Upon Request
[134] (2025)	strawberries/phenotyping, variety identification	RGB images, depth, near-infrared, MSI images (green, red, red-edge, near-infrared), thermal infrared, pixel temperature matrix, LiDAR point cloud data	Fast-SAM algorithm (RGB image segmentation; ROI masks); NDVI; NDWI; NRCT; P1; P2	early fusion	Calibrate intrinsic/extrinsic parameters and use transformation matrices to standardize camera data within the LiDAR coordinate system and generate fused point cloud data	Fast-SAM model	camera calibration toolbox in MATLAB (MathWorks, R2023a, USA) ; calibration plate method; targetless calibration approach; manual technical calibration; alignment of the projected image with the original image continuously monitored and adjusted	Canopy width: $R^2 = 0.9864$, RMSE = 0.0185 m. Average temperature: $R^2 = 0.8324$, RMSE = 0.1732 °C, Variety identification (clustering): ARI = 0.94 (NDVI + NDWI + NRCT + crown width + P1 + P2)	NR	NR	2894 ms/plant/-
[135] (2025)	Tobacco plant/plant stress assessment	EIS (Frequency, Impedance Magnitude, Impedance Phase), Temperature, RH, VPD, Volumetric Water Content (VWC), Plant Weight	NR	early fusion	unified multimodal input to AdapTree	AdapTree (Adaptive Boosted Tree ensemble combining AdaBoost and decision trees)	NR	Impedance: $R^2 = 0.993$, MAE = 22.789, RMSE = 134.565, RH: $R^2 = 0.999$, MAE = 1.51×10^{-5} , RMSE = 0.006966, Temperature: $R^2 = 0.999$, MAE = 2.51×10^{-5} , RMSE = 0.0050099	NR	NR	796,830 data samples/Upon Request
[136] (2025)	pepper/aphid early prediction	temperature, RH, light intensity, carbon dioxide concentration, number of aphids, aphid strain rate	maximum value, minimum value, average value, 1D CNN extracts the features of environmental factors, maximum pooling	data level fusion, feature-level fusion	weighted average fusion algorithm (primary fusion); heterogeneous sensor fusion algorithm (secondary fusion)	1D CNN-LSTM	NR	total RMSE = 1.503 improved by 6.891, 7.513, 33.980 (vs. 1D CNN, LSTM, BP); 1D CNN-LSTM—number of aphids (RMSE: 1.378, MAE = 0.900, $R^2 = 0.999$), aphid strain rate (RMSE = 0.337, MAE = 0.260, $R^2 = 0.999$)	CPU: Intel(R) Core(TM) i5-9300H processor, GPU: GeForce GTX 1650 graphics card, CUDA version 10.2; Microcontroller Unit (MCU), LoRa technology, WiFi module, OneNet cloud platform	NR	-/included in the manuscript

NR: Not reported.

6. AI- and IoT-Enabled Intelligent Greenhouse Systems

The Internet of Things (IoT) constitutes the core digital infrastructure of smart greenhouse systems by interconnecting sensors, imaging devices, communication modules, gateways, computing platforms, and actuators within an integrated monitoring and control architecture. In IoT-based SGHs, real-time data on key greenhouse variables—such as air temperature, relative humidity, light intensity, CO₂, and, where relevant, substrate or soil conditions, together with camera-based observations—are acquired and transmitted through communication networks for visualization, decision support, and automated actuation. This connectivity layer is essential because it enables continuous monitoring, remote supervision, and closed-loop control [137,138].

The integration of artificial intelligence with IoT infrastructures has further transformed smart greenhouses into data-driven production systems. Heterogeneous sensing streams, including microclimate, substrate and water variables, and imaging data, can be processed at the edge, in fog layers, or in the cloud according to latency, bandwidth, and energy constraints. This distributed computing paradigm helps overcome the limitations of cloud-only architectures while preserving scalability, and it has encouraged deployment-aware learning strategies that remain feasible under constrained computational resources.

From a control and decision-support perspective, AI translates sensor streams into actuation policies. For instance, adaptive particle swarm optimization combined with artificial neural networks has been applied for irrigation and fertilization management [139], whereas multilayer perceptron-based forecasters coupled with fuzzy logic have been employed to mitigate frost risk in tomato production [140]. Similarly, regression-based approaches using support vector machines or multilayer perceptrons provide accurate routine actuation under varying environmental conditions [141], while lightweight convolutional neural networks optimized through metaheuristic strategies have been demonstrated to detect operational anomalies directly at the edge without relying on continuous cloud connectivity [142].

For plant monitoring and production operations, vision-enabled IoT nodes complement scalar sensors by providing spatial information on crop health and phenotyping. Specifically, convolutional neural network-based classification combined with instance segmentation has been utilized to estimate fruit ripeness stages [143]. Extending this approach, visual detection models integrated with lightweight temporal predictors have been adopted to forecast harvest dates, demonstrating the combination of perception and temporal modeling [144]. In early-stage growth monitoring, RGB-D and multi-sensor setups coupled with advanced deep learning architectures enable detection of growth points and estimation of seedling height, highlighting the need for edge-cloud partitioning when three-dimensional perception increases computational demand [145]. For longer-horizon crop-state prediction, dynamic Bayesian networks have been employed to forecast leaf area index and evapotranspiration [146], while simpler learners such as k-nearest neighbors remain effective when the feature space is dominated by routine nutrient and water indicators [147].

Table 13 consolidates the reviewed AI and IoT systems in smart greenhouses with respect to sensing modalities, communication architecture, and reported performance, further details are presented in Table A6 provided in Appendix A.

Table 13. Overview of AI and IoT implementations in smart greenhouses.

Ref./Year	Crop	AI Task/Categorie	Architecture	Performance	IoT Communication Technology	Sensors	Data Type	Inputs	Target	Processing Level	Dataset	Dataset Availability
[139] (2022)	Olive	Irrigation, Fertilization decision/Classification	adaptive PSO-ANN	Acc = 94.8%, Pr = 91.15%, R = 97.93%, F1 = 94.42%, MAE 3.91	LM35, hygrometer, OMC-118, DHT22	time-series	NR	Olive type, temperature, soil moisture, wind, humidity	irrigation and fertilization	local control	MNIST, NSL-KDD, Syngenta Crop Challenge 2017, University of Arkansas plant dataset	NR
[140] (2020)	Tomato	Frost forecasting (air temperature)+control/Hybrid	MLP-ANN (BP training)	effectiveness > 90%, R ² = 89.27–95.22%	XBee-WiFi; GSM/GPRS (M2M); TCP/IP (via WAP)	Weather station (SparkFun Weather Shield DEV-12081) + wind/rain sensors (external)	time-series	ANN: Outside air temperature, Outside air relative humidity, Wind speed, Global solar radiation flux, Inside air RH Fuzzy: Predicted ANN temperature, Cropland temperature	ANN: inside air temperature Control target: pump activation (%PWM)	NR	Edge+Web	NR
[141] (2022)	Gerbera, Broccoli	Greenhouse control	SVM regressor, MLP (ANN) regressor	RMSE(avg): SVM = 0.11, MLP = 0.08; reported accuracy = 92%	Wi-Fi + MQTT (TCP/IP publish–subscribe) to Adafruit IO Cloud; serial to PC	DHT11, LDR, MQ2	On/off time duration of pump, ventilation fan, amount of light	1024 samples per sensing parameter; 70/30 train-test split	10 days	NR	NR	NR
[142] (2023)	Greenhouse vegetables (tomato, onion, peas, etc.)	Greenhouse monitoring	DCNN (Dilated CNN) + Fire Hawk Optimizer (FHO)	Accuracy: 95%;	NR	DHT11, YL69, LDR/light sensor, smoke sensor, water level sensor (with fan/pump/bulb/LCD)	NR	Temp, humidity, soil moisture, light, smoke, water level	Normal vs. out-of-range condition + control actions (fan/pump/light/alarm)	Edge (Raspberry Pi) + Cloud	NR	NR
[143] (2022)	Tomato	Disease detection/classification + Fruit ripeness monitoring (instance segmentation/detection)	CNN; Mask R-CNN	Avg accuracy = 0.91; weighted = 0.93	IP network (IP camera); Arduino–Raspberry Pi via USB serial; socket-based transfer to server	DHT11, moisture sensor, water-flow sensor, IP camera	Time-series + RGB images	Leaf/fruit images (256 × 256); sensor readings	Leaf disease class (10 diseases + healthy) + fruit ripeness stage (3 classes)	Edge (Raspberry Pi)/Server	PlantVillage tomato diseases: 8500 imgs; fruit stages: 1500 imgs (10,000 total)	Public
[145] (2022)	Watermelon + Pumpkin seedlings (greenhouse)	growth-point detection + height estimation	EfficientNet (BiFPN)	Growth-point detection: AP = 96.6%, F1 = 94%, time = 0.026 s; Height: R ² = 0.92–0.97, RMSE = 2.81–4.83 mm	4G cellular module (upload to cloud); TCP/IP video stream; Wi-Fi hotspot (PC↔Raspberry Pi)	Intel RealSense D415 + Azure Kinect + surveillance camera; BH1750, SHT30, CCS811; Silan A1 LiDAR; 9-axis IMU	RGB-D images + env time-series + video	RGB-D seedling images + light, Temperature, RH, CO2	Seedling height	Edge (Raspberry Pi/STM32) + Cloud server	Labeled growth-point images: 1600 (after augmentation), split 90/10 train/test; test set mentioned: 160 images	On request
[146] (2020)	Micro-tomato	Prediction/Regression	DBN (Dynamic Bayesian Network + EM)	Leaf Area Index (LAI): avg error 15.5%/12.2%/19.7% (Env1/2/3) and forecasts up to 21 days ahead; Evapotranspiration (ET) avg error 29.42%	Netsens Wireless Unit, Scale/weighing balance, Planimeter: DT Area Meter MK2	Time-series	NR	Growing Degree Days (GDD) (from temperature), Solar irradiance (R)	Leaf Area Index (LAI) and Evapotranspiration (ET)	NR	COLTIV@MI mini-greenhouse	NR

Table 13. Cont.

Ref./Year	Crop	AI Task/Categorie	Architecture	Performance	IoT Communica- tion Technology	Sensors	Data Type	Inputs	Target	Processing Level	Dataset	Dataset Availability
[147] (2023)	Leafy vegeta- bles	Crop growth management/Growth- rate prediction	KNN (K- Nearest Neighbor)	Acc = 93% (Coconut fiber + NFT: 93.3%); weighted avg 93%	NR	NR	Time-series sensor data + images	pH, EC, water temp, ambient temp, air temp/humidity	Crop growth rate/yield condition (leafy vegetables)	Cloud storage + offline analysis	Data from Univer- sity of Agricultural Sciences (GKVK), Bengaluru; 70/30 train-test split	NR
[148] (2023)	Tomato	Disease detec- tion/Classification	CNN	CNN: >90% overall; 92% validation accuracy, per- class precision/recall mostly 0.96–0.99 IoT: 99% successful trans- missions; max packet loss 2% and 44% energy reduction vs. no-sleep transmission	ZigBee/IEEE 802.15.4 (XBee S2C mesh) for sensor nodes; Wi-Fi for camera nodes; cellular interface from gateway to cloud	AM2315, SHT-10, SEN-08942, AM2302, OV2640 camera (ESP32-CAM)	time-series + RGB images	RGB leaf image (size: 200 × 200) env: air temperature, air hu- midity, soil moisture, soil temperature	Tomato leaf health state (healthy vs. dis- ease class; 10 tomato categories)	Edge/Cloud	tomato subset used: 16,012 im- ages (12,810 train/ 3202 validation)	PlantVillage: public, IoT data: On request

NR: Not reported.

7. Current Challenges and Research Gaps in Smart Greenhouse Applications: Opportunities and Improvement Strategies

Supervised AI models depend strongly on the quality of the training dataset. In the realm of SGH's research, there is a lack of publicly standardized benchmark datasets. Furthermore, the existing ones are associated with one or more of the following limitations:

- Small-scale datasets: The utilized datasets are often cropped in a short time, such as a limited season, containing a small number of images or short sensor logs.
- Single-modality: Many datasets contain either vision data, such as leaf/fruit pest images, or sensor data such as temperature, humidity, CO₂
- Restricted crop and stage diversity. Many datasets only focus on one crop or a narrow window of phenological stages, limiting the variation in cultivar, canopy structure, and appearance of symptoms.
- Insufficient temporal coverage: Sensor datasets are often limited to a single growing cycle and therefore only represent a subset of the environmental variability, increasing their risk of seasonal drift and unusual occurrences.

To tackle these limitations, researchers rely on data augmentation techniques, transfer learning models, and Synthetic data generation models. Data augmentation expands the effective training distribution by simulating plausible variations in geometry, illumination, noise, and occlusion. In SGH robotic harvesting, ref. [105] employed geometric and photometric transformations together with occlusion simulation, demonstrating enhanced robustness under occlusion and strong detection performance, with mAP@0.5 reported as 0.947. Similarly, ref. [149] used simple augmentation strategies and reported substantial gains in recall and F1-score after augmentation and enhancement. These findings indicate that augmentation remains a practical and effective strategy when collecting additional real greenhouse data is difficult. In parallel, transfer learning is frequently adopted when SGH datasets are too limited to support deep network training from scratch. By initializing models with weights pretrained on large-scale datasets such as ImageNet, it becomes possible to reuse low-level and mid-level features, thereby improving downstream performance. For example, ref. [150] fine-tuned a pretrained YOLOv9 pipeline for pest and disease imagery. Likewise, refs. [105,149] reported improved outcomes when leveraging pretrained weights in harvesting and aquaponic detection tasks. Taken together, these studies suggest that transfer learning is an effective means of compensating for limited SGH-specific data availability. Synthetic data approaches further aim to reduce labeling costs and improve coverage of challenging conditions, such as class imbalance, extreme occlusion, and variations in illumination. For example, ref. [151] used SMOTE to address minority-class actuator states. For greenhouse pest imagery, Karam et al. [152] combined Copy-Paste-Blend with GAN-generated masks to synthesize realistic infestations across crops, reporting improved detection performance when training data were limited. For heavy occlusion scenarios, Son et al. [153] generated condition-controlled synthetic occlusions and illumination variants to train models that are more robust to canopy clutter, again reporting improved performance under challenging settings. Accordingly, synthetic data generation appears to be a promising complementary strategy for addressing data scarcity in SGHs, particularly when rare or difficult cases are underrepresented in real datasets.

Beyond dataset scarcity, a major challenge lies in the limited generalization of models across greenhouse environments. Machine learning models trained in one greenhouse often degrade when deployed in another due to differences in cultivar, camera optics, illumination, substrates, layout, and management practices. Despite the practical importance of this issue, external validation across multiple sites remains uncommon, and domain adaptation methods are not yet sufficiently mature for SGH-specific conditions. As a result, cross-site robustness continues to represent a key obstacle to large-scale deployment.

Another important challenge concerns forecasting reliability under imperfect sensing conditions. In practical SGH deployments, sensor streams frequently contain missing values due to communication dropouts, calibration drift, and sensor faults. Many studies address missing data using simple ad hoc preprocessing steps, such as interpolation or forward filling; however, they often omit the missing-data rate and do not evaluate how the selected imputation method affects forecast accuracy. To improve robustness, future forecasting pipelines should treat missingness as part of the learning problem by reporting missing-data statistics, describing missingness patterns, and adopting models that either perform imputation jointly with forecasting or explicitly account for sensor reliability. Greenhouse-oriented imputation approaches provide a strong foundation [154], and edge-side imputation has also been explored to prevent corrupted measurements from propagating through the pipeline [155]. Therefore, handling missingness explicitly is not only a preprocessing concern but also a central requirement for reliable forecasting in operational SGHs.

In addition to data and forecasting issues, perception in greenhouse environments remains particularly challenging under overlapping structures, occlusion, clutter, and small targets. In real greenhouse imagery, tomato disease and pest symptoms are frequently affected by lighting variation, shadows, branch or leaf occlusion, and dense leaf overlap, making small targets easy to miss and reducing detection performance. To address these issues, Wang et al. [156] enhanced YOLOv3-tiny with an inverse-residual block (IRB) backbone, together with foreground-region training and multi-scale strategies. In their comparison, YOLOv3-tiny-IRB achieved mAP = 93.1% and F1 = 0.922, outperforming YOLOv3-tiny (88.1%, 0.893) and YOLOv3 (88.8%, 0.897), as well as other baselines such as SSD, Mask R-CNN, and Faster R-CNN, while maintaining real-time speed. These results highlight the importance of designing perception models that explicitly account for greenhouse-specific visual complexity.

Amodal segmentation also represents a relevant direction for handling hidden structures in SGH imagery, as it seeks to recover object extent beyond visible regions and is therefore well aligned with fruit and organ delineation under occlusion. Although promising, amodal methods remain underrepresented in SGH benchmarks and real deployment reports. Sungjay Kim et al. [157] addressed heavy leaf and fruit occlusions by training an amodal instance segmentation pipeline built on Mask R-CNN combined with a reconstruction network, first tested with an autoencoder and later with a U-Net reconstruction network, using synthetic leaf patches to simulate occlusion. Their best U-Net variant achieved AP = 50.06 and AP50 = 82.43 with an inference time of approximately 220 ms per image. Jing Yang et al. [158] proposed a Transformer-based amodal segmentation model (ACBET) using a Swin Transformer U-Net backbone with boundary estimation and a GAN-based reconstruction loss, trained largely from modal masks to reduce the cost of amodal annotation. On a dataset collected from two greenhouses, they reported 96.07% average pairwise accuracy, 94.13% mIoU, and 57.79% invisible mIoU for recovering hidden regions. Moreover, by using the generated pseudo-amodal masks to train Mask R-CNN, they obtained AP = 63.91% and AP50 = 86.91%. Collectively, these findings suggest that amodal segmentation can provide a useful direction for improving perception under severe occlusion in SGHs.

Related to this, condition-controlled synthetic occlusions provide targeted training samples that better reflect canopy clutter in greenhouse scenes. Son et al. [153] illustrated this approach by generating structured occlusion ratios and lighting variants. Similarly, GAN-assisted synthesis can densify crowded scenes for tiny-pest detection [152]. In this respect, synthetic occlusion modeling complements conventional augmentation by enabling more controlled exposure to difficult visual conditions. Likewise, small-object detection in

cluttered scenes often benefits from multi-scale feature fusion and attention mechanisms embedded within detector backbones and heads. Such enhancements are particularly relevant in SGHs, where pest symptoms, fruit structures, and leaf abnormalities may occupy only a small fraction of the image and may appear under strong background interference.

Finally, beyond perception accuracy, SGH monitoring and control often require low-latency inference under constrained computation, limited power budgets, and intermittent connectivity. Energy-aware scheduling, which jointly optimizes sensing, communication, and inference, is also becoming increasingly relevant for sustainable deployment. Consequently, practical SGH AI systems must be evaluated not only in terms of predictive performance but also with respect to their deployability under real-time and resource-constrained operating conditions.

8. Future Research Directions

To address the gaps above, future SGH research should prioritize several complementary directions. Establishing standardized open datasets is essential. High-value benchmarks should include multiple crops, cultivars, phenological stages, camera types, and microclimates, collected across seasons and sites to enable robust external validation. In parallel, synthetic data generation, such as GANs, can help address class imbalance and rare conditions, but its utility must be demonstrated rigorously through (i) realism assessment, (ii) controlled ablations, and (iii) deployment tests that quantify synthetic-to-real transfer.

Another important priority is the development of fusion-ready multimodal datasets that provide synchronized imagery and sensor data, including temperature, humidity, CO₂, light, and irrigation, together with consistent timestamps and reliable ground truth. Such resources would support context-aware monitoring and promote more stable performance under visual noise and domain shift. At the same time, greater attention should be given to deployment-oriented AI systems tailored to embedded hardware and real operational constraints, including limitations in computation, power, and connectivity. Beyond initial deployment, research should also consider the full model lifecycle, including monitoring for drift, safe model updates, and calibration under changing operational conditions.

In this context, federated learning represents a promising strategy for collaborative model training across multiple greenhouses without centralizing raw data, thereby potentially improving cross-site generalization under privacy and bandwidth constraints. Likewise, active learning offers a practical mechanism for reducing annotation cost by prioritizing the most informative samples for labeling. Although both paradigms show clear potential for SGH AI, systematic investigations under real operational greenhouse conditions remain limited, highlighting a significant research opportunity. Finally, future work should also aim to balance predictive accuracy with reliable communication and low-latency actuation in IoT-enabled SGHs. In such settings, models intended for edge deployment may benefit from compression, quantization, and distillation to support efficient and real-time execution while maintaining practical utility.

9. Discussion

This review synthesizes AI-enabled SGH studies across different data modalities, including environmental time-series, RGB, and spectral sensing. In doing so, it clarifies how methodological choices in the reviewed studies are shaped by sensor availability, task definition, and deployment constraints [36,67,113,131].

Table 14 shows that prior smart-greenhouse reviews are mostly fragmented, focusing on subsets such as microclimate modeling and optimization, control strategies, and time-series forecasting or deep-learning computer vision. In contrast, our 2026 review delivers an

end-to-end synthesis that unifies sensing (deep-learning computer vision and multispectral, hyperspectral, or thermal sensing), multimodal data fusion, microclimate modeling and optimization, time-series forecasting, control strategies, and edge or embedded deployment with robotic harvesting, providing a clearer roadmap for deployable autonomous smart-greenhouse systems.

Across the reviewed studies, crop coverage is imbalanced. Most of the studies focus on tomatoes, while fewer ones address peppers, cucumbers, and other crops [32,39,47]. Within climate-focused research, temperature is the most frequently modeled variable, followed by relative humidity, which is consistent with their central role in greenhouse operation and their routine availability in sensor networks. This trend is evident from the dominant targets reported in Table 2 [32,36,39,40,46].

A key difficulty in evidence synthesis is the limited comparability across studies. The reviewed papers differ in targets, sensing configurations, acquisition settings, and evaluation protocols, and they often report different metrics. In addition, computational efficiency is not consistently reported in terms of inference time, and model size, which weakens the strength of cross-study comparisons. This issue is visible in the uneven reporting of deployment-related fields in [36,68,71].

Regarding model families, the reviewed evidence supports a cautious and task-dependent interpretation. Deep learning is frequently selected for high-dimensional perception problems and for strongly nonlinear dynamics, while classical machine learning and ensemble approaches remain competitive when data are limited, inputs are structured, or efficiency and interpretability are prioritized. This is supported by the continued use of SVR and gradient-boosting methods for microclimate prediction in Table 2 [32,38–40]. Moreover, ensemble and hybrid pipelines appear repeatedly in the reviewed studies, indicating an emphasis on robustness through complementary predictors, including stacking ensembles and process-guided hybrid models [41,45,46,48].

For computer-vision applications, deep learning dominates the reviewed SGH literature. YOLO-family detectors are widely used because they provide a practical balance between accuracy and throughput, as shown by the many detection-oriented entries in the summarized tables. At the same time, the reviewed studies also show that performance is sensitive to greenhouse-specific factors such as small targets, occlusion, variable illumination, and dataset and annotation choices, which motivates architectural modifications and task-specific training strategies [67,68,70,88,93].

At the sensing level, RGB imaging is the most widely used modality in the reviewed studies due to its low cost and ease of acquisition, including smartphone-based collection, for tasks such as disease recognition, pest monitoring, harvesting perception, and growth-stage assessment [70,71,88]. In contrast, MSI and HSI provide richer spectral information and can support stress detection and earlier disease discrimination by capturing reflectance changes in red-edge and near-infrared regions [113–116]. However, the reviewed evidence also highlights practical constraints for spectral pipelines, including calibration demands and sensitivity to sensor properties, illumination conditions, acquisition protocols, and plant surface properties [114,115,117].

Finally, the reviewed studies indicate that multimodal data fusion can improve robustness by combining complementary information sources such as imagery with environmental context. Table 12 summarizes representative fusion designs and performance gains, including feature-level fusion for disease detection, sensor-level fusion for stress assessment, and temporally aware fusion for early pest warning [131,133,135,136]. In the reviewed SGH settings, common contextual signals include temperature and humidity, as well as soil moisture and light intensity, which can support earlier and more stable inference when combined with vision [32,39,40,46,55].

Table 14. Comparison of smart-greenhouse review papers. MMO: microclimate modeling/optimization; CTRL: control strategies; TSF: time-series forecasting; CV: deep-learning computer vision; SPEC: multispectral/hyperspectral/thermal; IOT: IoT/WSN; DATA: multimodal fusion; EDGE: edge/embedded deployment; ROB: robotics harvesting.

Ref.	Year	MMO	CTRL	TSF	CV	SPEC	IOT	DATA	FUS	EDGE	ROB
[23]	2021	✓	✓	✓	×	×	×	×	×	×	×
[18]	2020	✓	✓	✓	×	×	✓	×	×	×	×
[19]	2024	✓	×	✓	×	×	×	×	×	×	×
[20]	2024	×	×	×	✓	×	×	✓	✓	✓	×
[22]	2025	×	✓	×	×	×	✓	✓	×	×	×
[21]	2025	✓	✓	✓	✓	×	✓	✓	✓	×	×
[24]	2025	✓	✓	✓	×	×	✓	×	×	×	×
[25]	2026	×	✓	×	×	×	✓	✓	✓	✓	✓
Ours	2026	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

10. Conclusions

This review synthesizes research progress on the integration of IoT and ML/DL using diverse data modalities in major smart greenhouse applications, including disease detection, pest and insect monitoring, yield estimation, plant growth analysis, and robotic harvesting. It highlights that data fusion approaches are emerging as a promising direction for improving model robustness and enabling the use of complementary information for more reliable decision-making. Despite these advances, several challenges persist. The lack of large-scale, publicly available, and standardized datasets limits model generalization, and this limitation is further compounded by the sensitivity of smart greenhouse models to site- and structure-specific characteristics, such as greenhouse geometry, construction typology, covering material, dimensions, and orientation. In addition, most proposed models remain insufficiently validated in real smart greenhouse environments, while computational constraints of embedded systems hinder real-time deployment. Furthermore, environmental conditions such as occlusion, object overlap, and missing data continue to affect model performance, and the limited use of synthetic data generation techniques represents an additional gap. Overall, although AI and IoT technologies demonstrate strong potential for enabling automation in smart greenhouses, further improvements are required to address these limitations.

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Abbreviations

The following abbreviations are used in this manuscript:

SGHs	smart greenhouses
ARX	autoregressive with exogenous input
ARIMA	autoregressive integrated moving average
SARIMA	seasonal autoregressive integrated moving average
AIC	the Akaike information criterion
ML	machine learning
SVM	support vector machines
MAPE	mean absolute percentage error
RNN	recurrent neural networks
LSTM	long short-term memory
GRUs	gated recurrent units
MLR	regression
RF	random forest
GBDT	gradient boost decision tree
XGBoost	extreme gradient boosting
SGB	stochastic gradient boosting
BP	back propagation
PCA	principal component analysis
VCPA	variable combination population analysis
IRIV	iterative retention of informative variables
CWSI	crop water stress index
TRI	thermal images
RH	relative humidity
T_a	air temperature
ViT	vision transform
EIS	electrical impedance spectroscopy
VPD	vapor pressure deficit

Appendix A. Detailed Tables of the Reviewed Studies

Table A1. Overview of architectures and resources for SGHs applications.

Ref.	Peroid	Papers Studied	Domain	Tasks	Objective	Finding
[16]	2021–2025	100	Intelligent and autonomous greenhouse systems	Environmental sensing; microclimate prediction/control; disease and pest detection; growth, yield, and quality assessment; irrigation/fertilization management; multi-sensor fusion; AI-based forecasti, robotic task execution	rovide a systematic technical review of AI, IoT, ML/DL, and robotics applications in SGHs management; evaluate environmental and crop control approaches; identify research gaps and propose future directions for autonomous greenhouse systems	- Concludes that deep-learning multimodal fusion improves disease and stress detection and enhances microclimate prediction relative to single-sensor approaches. - Emphasizes that IoT/WSN connectivity, including ZigBee, LoRa, and NB-IoT, enables real-time monitoring and more precise automation, supporting gains in resource efficiency and crop yield. - Notes that robotics can automate greenhouse operations and reduce manual labor, but adoption at large scale remains limited. - Identifies key challenges including heterogeneous real-time data fusion, limited interpretability and cross-site generalization, lack of sensor standardization, and cost and deployment barriers.
[18]	NR	NR	artificial neural networks (ANNs)	microclimate prediction/control, Energy optimization, CO ₂ enrichment management	Examined the use of artificial neural networks(ANNs) in greenhouse technology, discussing training methods and the pros and cons of optimization models	- Feedforward networks dominate; recurrent/hybrid ANNs are under-used - Optimization (GA/PSO, fuzzy) and data-reduction (PCA) improve ANN training/performance. - ANNs can outperform classic physical models for short-term indoor-climate prediction and run faster; hybrid ANN+physics is promising but still scarce. - DL/CNN/RNN and WSN/IoT are growth areas (disease detection, image-based microclimate, multi-greenhouse data sharing).
[19]	NR	61	ML and DL time-series forecasting	microclimate prediction, water-quality/dissolved-oxygen forecasting, energy/consumption and related environment variables (Temperature (32.6%) and dissolved oxygen (19.6%) are the most frequent targets)	Conduct an SLR that maps the development of time-series models for greenhouse environments, evaluates accuracy and generalization, analyzes parameter/time-step effects, and identifies frontier models (attention and GNN).	- Deep learning surpasses traditional/statistical ML for nonlinear, spatio-temporal greenhouse data; attention and memory modules (RNN/LSTM/GRU) substantively improve long-sequence prediction. - Shorter sampling intervals improve forecast accuracy for greenhouse time-series, while larger time steps degrade baseline models; attention-based architectures substantially mitigate this accuracy loss.

Table A1. Cont.

Ref.	Peroid	Papers Studied	Domain	Tasks	Objective	Finding
[20]	2020–2023	100	Deep learning enabled computer vision	Growth monitoring, disease detection, yield estimation, recognition/classification of crops, quality inspection, automatic harvesting, pest/insect monitoring, crop health analysis, Seed Quality Analysis, and Weed Management	Deliver a focused review analyzing capabilities and limitations, summarizing performance tables, and outlining challenges and future directions for CV/DL adoption. The techniques, datasets, models, and overall performance results reported in the literature are analyzed.	- Field faces data limitations (few large, labeled greenhouse datasets) and weak transfer from open-field pretraining; - need for lightweight models suitable for embedded/edge deployment; - multimodal fusion (RGB + depth/thermal/hyperspectral) is promising but underused; - talent and deployment gaps hinder real-world scalability; - overall, DL-CV shows strong potential for autonomous, efficient greenhouse operations if these gaps are addressed.
[21]	NR	NR	Sensing technologies for tomato greenhouse production; non-invasive monitoring; multi-sensor fusion; AI-based diagnostics	Environmental sensing, plant physiological sensing, microclimate prediction and control; disease/stress detection; growth/yield/quality assessment; irrigation/fertilization management; multi-sensor fusion frameworks; AI (ML/DL/RL, ensembles) for detection and forecasting.	Provide technical guidance by systematically examining key environmental factors, high-throughput non-destructive sensing technologies, multi-sensor fusion and data-driven diagnostic systems, and research gaps/future trends for intelligent tomato production.	- Deep-learning multimodal fusion enhances disease/stress diagnosis accuracy compared to single-sensor approaches; non-invasive sensing enables earlier detection before visual symptoms appear. - IoT/WSN integration (ZigBee/LoRa/NB-IoT) underpins precise, automated environmental control and can boost resource efficiency and yield
[22]	2020–2024	74	AI-driven SGHs	climate control, secure data transmission and storage, disease and pest detection, growth evaluation or big data management and synthesis.	Provide a state-of-the-art synthesis of AI and sensory data for SGHs, assess alignment with current AI trends, and bridge academic–industry gaps	- Data limitations prevail (few comprehensive, multi-season, multi-sensor datasets; lack of benchmarks), hindering generalization and comparability; data-centric practices are scarce. - Most works use limited sensor types and offline logs; multimodality and real-time + batch processing are under-adopted - ML/DL dominate; RL and emerging trends (e.g., LLMs, explainable AI, generative AI, metaverse/digital twins at scale) are weakly represented.
[23]	2019–2020	73	mechanism, time-series, machine learning	Microclimate prediction, Climate control/optimization	Reviewed greenhouse modeling technologies, including mechanistic, time series, and machine learning approaches.	- dynamic and neural network approaches dominate; deep learning (RNN/LSTM) adoption has risen in 2018–2020 - widespread use of heuristic/swarm algorithms (e.g., PSO, EA) for parameter identification and hyperparameter tuning - temperature and humidity are the most common targets (73% of papers)

Table A1. Cont.

Ref.	Period	Papers Studied	Domain	Tasks	Objective	Finding
[24]	2014–2023	107	Environmental control strategies (structural, parameter, algorithmic); greenhouse models (microclimate and crop-growth models)	Heating/cooling/lighting/ventilation control; PID, fuzzy, MPC, ANN and hybrid algorithms; mechanistic and data-driven microclimate models	Provide an overview of greenhouse control strategies and models, highlighting trends and future research directions	- Notes that PID remains widely adopted, while neural-network control and model-predictive control are increasingly dominant. - Highlights the use of heuristic and swarm-based optimizers for parameter identification and controller tuning. - Observes that most studies prioritize temperature and humidity as primary control targets. - Summarizes that microclimate modeling is mainly addressed through ANN-based predictors and CFD-based analysis.
[25]	2010–2025	106	Control strategies (conventional and intelligent) and modeling techniques (mechanistic, CFD, data-driven, crop-growth models)	Climate control (temperature, humidity, CO ₂ , light); irrigation and crop-growth control; neural network and reinforcement-learning-based controllers; integration of intelligent modeling with control	Systematically summarize conventional versus intelligent control strategies and modeling approaches and highlight research gaps and future challenges	- Reviews 106 papers, with 33 on conventional controllers such as PID, fuzzy control, and MPC, and 34 on intelligent controllers. - Breaks down intelligent control into neural-network-based methods, reinforcement-learning-based methods, and other AI-driven strategies. - Notes that earlier surveys rarely integrated advanced intelligent control with data-driven modeling in a unified synthesis. - Provides a structured overview of control strategies and modeling approaches, and highlights gaps in real-time adaptability and sustainability-oriented design.
NR: Not reported.						

Table A2. Summary of data-driven microclimate forecasting studies in SGHs.

Ref./Year	Crop	Category	Architecture	Performance	Sensors/Weather Station	Target	Interval	Outdoor Input Variables	Indoor Input Variables	Dataset	Period	Dataset Availability
[29] (2023)	NR	Hybrid modeling	ARTFIMA–SVM	Air temperature: RMSE = 0.79, MSE = 0.62, MAE = 0.44, MAPE Moisture: RMSE = 1.94, MSE = 3.79, MAE = 1.46, MAPE = 0.02	NR	Internal air temperature, Moisture	10 min	NR	Air temperature, Moisture	1987 samples	NR	Private
[32] (2023)	Tomato, bell pepper	Classical machine learning	SVR + polynomial kernel	Fall: $R^2 = 0.9808$, RMSE = 1.7431, MAE = 0.2856, MAPE = 0.0124 Winter: $R^2 = 0.9984$, RMSE = 0.6760, MAE = 0.2929, MAPE = 0.0123 Spring: $R^2 = 0.9924$, RMSE = 1.0362, MAE = 0.7444, MAPE = 0.0270 Summer: $R^2 = 0.9999$, RMSE = 0.0549, MAE = 0.04222, MAPE = 0.0015	Davis Vantage Pro 2 weather station; SHT11 temperature/humidity sensor	Internal air temperature	5 min	Air temperature, Air relative humidity, Solar radiation	Air temperature, Air relative humidity, Internal dew Point	85,989 samples	2020–2021	Public

Table A2. Cont.

Ref./Year	Crop	Category	Architecture	Performance	Sensors/Weather Station	Target	Interval	Outdoor Input Variables	Indoor Input Variables	Dataset	Period	Dataset Availability
[33] (2023)	Cucumber	Classical machine learning	RBF	RMSE = 1.32 C, MAPE = 3.23%, $R^2 = 0.931$	AM2303 digital temperature and humidity sensors, TES132 solar power meter, DT186 anemometer	Internal air temperature	5 min	Air temperature, Air relative humidity, Solar radiation	Air temperature	NR	1 month (2020)	On request
[34] (2023)	NR	Classical machine learning	RBF + Levenberg–Marquardt	RMSE = 0.91, MAPE = 1.30	SHT11 temperature/humidity sensors; TES1333R solar meter; wind speed from Soda Service	Internal air temperature	5 min	Solar Radiation, Air Temperature, Air relative humidity, Wind Speed	NR	NR	2022 (10 days)	On request
[36] (2021)	NR	Shallow Neural Networks	ANN	RMSE = 0.289–0.402, MAPE = 0.87–1.04%, $R^2 \geq 0.997$	NR	Internal air temperature	10 min	NR	Air temperature	40,033 samples	08/2019–11/2020	On request
[37] (2022)	NR	Deep sequential learning	GRU	$R^2 = 0.87$, RMSE = 3.58	Thermal sensor, National Meteorological Information Center of China (NMIC)	Minimum Internal air temperature	1 h	Air temperature, Maximum air temperature, Minimum air temperature, Rainfall, Air pressure, Vapor pressure deficits, Dew point temperature, Air relative humidity, Minimum air relative humidity, Two-minute average wind speed, Two-minute average wind direction, Occurrence of ground maximum temperature, Ground minimum temperature	NR	NR	20/2017–30/2018	On request
[38] (2023)	NR	Deep sequential learning	Seq2seq GRU with Luong attention	Air temperature: MAE = 0.228760, RMSE = 0.274486, $R^2 = 0.626864$. Realtime humidity: MAE = 1.420797, RMSE = 1.666903, $R^2 = 0.843962$	NR	Air temperature, Air relative humidity	NR	NR	NR	273,144 timesteps	NR	On request
[39] (2023)	Cucumber	Deep sequential learning	FAM-LSTM	MAE 0.6 °C, RMSE 0.9 °C (12 h); MAE 0.8 °C, RMSE 1.1 °C (24 h); MAE 1.4 °C, RMSE 1.7 °C (36 h); MAE 1.7 °C, RMSE 2.3 °C (48 h). Humidity (split 8:2): MAE 0.8%, RMSE 1.0% (12 h); MAE 2.5%, RMSE 3.1% (24 h); MAE 2.6%, RMSE 3.7% (36 h); MAE 2.5%, RMSE 3.5% (48 h)	JXBS3001, LH-SL1, LH-LR2000, LHGC2000	Internal air temperature, Air relative humidity	5 min	Air temperature, Soil air temperature, Air relative humidity, Soil relative humidity, CO ₂ concentration	Air temperature, Soil air temperature, Air relative humidity, Soil relative humidity, Light intensity	23,344 samples	2021–2022	On request

Table A2. Cont.

Ref./Year	Crop	Category	Architecture	Performance	Sensors/Weather Station	Target	Interval	Outdoor Input Variables	Indoor Input Variables	Dataset	Period	Dataset Availability
[40] (2022)	Cucumber	Ensemble learning (Boosting)	LightGBM (LGBM)	RMSE = 0.645	Priva sensors	Internal air temperature	5 min	Air temperature, Solar radiation	Air relative humidity, Soil temperature, Soil moisture, CO ₂ concentration, Aperture of outside shading curtain, Aperture of inside shading curtain, Aperture of inside energy-saving curtain, Aperture of ventilation window, Output air temperature of ATUs	495,714 samples	2014–2019 (2016 excluded)	Public
[41] (2023)	Tomato	Hybrid modeling	PBI + PF + DNN	temperature: RRMSE 6.7%, vapor-pressure: RRMSE 12.6%; seasonal total fresh-yield error 0.9%, cumulative-yield RRMSE 6.6%.	NR	Internal air temperature, Internal vapor pressure, Seasonal fresh yield prediction	5 min	Global irradiation, Air temperature, Air vapor concentration, CO ₂ concentration, Wind speed, Sky temperature, External soil-layer	Air temperature, Air relative humidity, CO ₂ concentration	NR	2015–2020	Private
[45] (2023)	Cherry tomato	Hybrid modeling	DF-RF-ANN	Air temperature: $R^2 = 0.72$, MAE = 1.97, Air relative humidity: $R^2 = 0.68$, MAE = 0.10, PAR: $R^2 = 0.75$, MAE = 76.39, CO ₂ concentration: $R^2 = 0.59$, MAE = 10.14,	STMAS-WRF (CWA)	Internal air temperature, Air relative humidity, Photosynthetically active radiation (PAR), CO ₂ concentration	10 min	surface temperature, Vapor pressure deficit, Dew point temperature, Long-wave radiation, Surface pressure, Air relative humidity, atmospheric pressure, and short-wave radiation	Air temperature, Air relative humidity, Photosynthetically active radiation (PAR), CO ₂ concentration	NR	01/04/2020–13/07/2021	On request
[46] (2024)	Cherry tomato	Hybrid ensemble learning	PSO-BiGRU-Attention-LightGBM	30 min: Air Temperature $R^2 = 0.9902$, RMSE = 0.5578, MAE = 0.3172; Air relative humidity $R^2 = 0.9825$, RMSE = 2.2327, MAE = 1.1981; PAR $R^2 = 0.8871$, RMSE = 39.1562, MAE = 18.4348. 60 min: Air Temperature RMSE = 0.7936, MAE = 0.4793, $R^2 = 0.9803$; PAR: RMSE = 46.2710, MAE = 22.8484, $R^2 = 0.8423$; Air relative humidity: RMSE = 3.1482, MAE = 1.7270, $R^2 = 0.9652$, 120 min: Air Temperature $R^2 = 0.9586$; Air relative humidity $R^2 = 0.9232$; PAR $R^2 = 0.8066$.	JXBS-3001, small weather station QI	Air temperature, Air relative humidity, Photosynthetically active radiation PAR	10 min	Air temperature, Air relative humidity, Photosynthetically active radiation PAR, Wind speed	Air temperature, Air Relative humidity, Photosynthetically active radiation PAR, Light intensity, CO ₂ concentration	37,008 samples	23/09/2020–06/06/2021	On request
[47] (2023)	Tomato	Deep sequential learning	LSTM-RNN	$R^2 = 0.80$, RMSE = 9.13%, MAE = 4.45%, MaxAE = 62.94%	Campbell Scientific HC2S3, Hukseflux LP02, E+E Elektronik EE820-C2, Vector Instruments A100L2/PC3	Vent opening	30 s	Air temperature, Air relative humidity, CO ₂ concentration	Solar radiation, CO ₂ concentration, Air relative humidity, Air temperature, Difference between inside and outside air relative humidity, Difference between inside and outside CO ₂ concentration, Difference between inside and outside air temperature, Vent opening signal	233,280 samples	(10–29)/12/2020	On request

Table A2. Cont.

Ref./Year	Crop	Category	Architecture	Performance	Sensors/Weather Station	Target	Interval	Outdoor Input Variables	Indoor Input Variables	Dataset	Period	Dataset Availability
[48] (2024)	Ornamental plants	Ensemble learning (Stacking)	Stacking DT + RF + KNN + XGB	$R^2 = 0.96515$, MAE = 0.01395, MSE = 0.03205, RMSE = 0.00102.	DHT22 temperature and humidity sensor, MQ7 carbon monoxide sensor, Light-dependent resistance (LDR)	Internal air relative humidity	15 s	Air temperature, Air relative humidity	Air relative humidity, Air temperature, CO saturation, Luminosity, Temperature difference (indoor–outdoor) feature	604,135 samples	08/2021–12/2021	On request
[49] (2023)	NR	Deep sequential learning/ensemble learning (boosting)	LSTM-RNN, XGBoost	LSTM-RNN best result in summer: $R^2 = 0.9994$, RMSE = 0.2698, MAE = 0.1449, MAPE = 0.0041; XGBoost best result in winter: $R^2 = 0.9345$, RMSE = 4.4443, MAE = 2.7693, MAPE = 0.0809	NR	Air temperature	NR	Air temperature, Air relative humidity, solar radiation	Air relative humidity, Dew point, Air temperature	NR	07/2020–06/2021	Public
[50] (2025)	Tomato	Deep sequential learning	PLSTM	$R^2 = 0.9999$, RMSE = 0.0220, MAE = 0.0160	NR	NR	Internal air temperature	NR	Air temperature, Solar radiation, Air relative humidity	Air relative humidity, Dew point	15/02/2018–14/08/2018	On request
[51] (2025)	NR	Hybrid modeling	LSTM-SVM	Air temperature (LSTM best): RMSE = 0.0766, MAE = 0.0454, $R^2 = 0.8825$; Air relative humidity (best overall): RMSE = 5.3034, MAE = 3.8041, $R^2 = 0.8187$; Classification (SVM): Accuracy = 0.63, Precision = 0.64, Recall = 0.63, F1-score = 0.63;	DHT11, ESP32 microcontrollers; ESP-NOW used for low-latency wireless	Air temperature, Relative air humidity, Environmental states	10 min	Air temperature, Air relative humidity	Air temperature, Air relative humidity	NR	NR	Private
[52] (2024)	Tomato	Deep sequential learning	DLinear, SegRNN	1 h—DLinear (Air temperature): $R^2 = 0.938$, RMSE = 0.293, MAE = 0.189; (Air relative humidity): $R^2 = 0.857$, RMSE = 0.427, MAE = 0.273; SegRNN (CO ₂ concentration): $R^2 = 0.875$, RMSE = 0.371, MAE = 0.231. 3 h—DLinear: -(Air temperature): $R^2 = 0.833$, RMSE = 0.479, MAE = 0.312; (Air relative humidity) $R^2 = 0.680$, RMSE = 0.640, MAE = 0.458; SegRNN (CO ₂ concentration): $R^2 = 0.711$, RMSE = 0.552, MAE = 0.354	SH-VT260, Vantage Pro2	Internal air temperature, Internal air relative humidity, CO ₂ concentration	NR	Air temperature, Air relative humidity, Wind speed, Wind direction	Air temperature, Air relative Humidity, CO ₂ concentration, Fan, Fogging, CO ₂ injection, Window openness, Shade curtain, Heat retention curtain	6744 samples	22/09/2020–29/06/2021	On request
[53] (2025)	Tomato	Hybrid modeling	GWO-BiLSTM	10 min: $R^2 = 0.97$, RMSE = 0.7889, MAPE = 4.94, MSE = 0.63, MAE = 0.62; 30 min: $R^2 = 0.97$, RMSE = 0.8863, MAPE = 8.5, MSE = 0.68, MAE = 0.65	PH-CJ1; BNL-GPRS-10G; GSP-6	Internal air temperature	5 min	Air temperature, Air relative humidity, Illumination, Wind speed, Wind direction	Air relative humidity, Light, CO ₂ concentration	124,837 samples	13/09/2024–1/11/2024	On request

Table A2. Cont.

Ref./Year	Crop	Category	Architecture	Performance	Sensors/Weather Station	Target	Interval	Outdoor Input Variables	Indoor Input Variables	Dataset	Period	Dataset Availability
[54] (2025)	NR	Classical machine learning	RF, MLR	Sunny: RF: $R^2 = 0.78$, RMSE 1.37 °C, MAPE 4.54%, Cloudy: RF: $R^2 = 0.81$, RMSE 1.33 °C, MAPE 4.14%, Overcast: MLR: $R^2 = 0.82$ (five-fold CV RMSE 0.68 °C); RF lower at $R^2 = 0.72$.	RC-4HC, RC-4	Internal air temperature	1 h	Air temperature, Air relative humidity, Ground wind speed	Air temperature, Water temperature	NR	11/2024–01/2025 + 2021–2025	Private
[55] (2025)	NR	Hybrid modeling	RIME-CNN-BiLSTM	Air temperature: MAE = 0.598, RMSE = 0.826, $R^2 = 0.977$. Air relative humidity: MAE = 1.698, RMSE = 2.287, $R^2 = 0.981$	PTS-3, PTWD-2A, TDR-3, RS-WS-1, P33-C49, TBQ-6, TBQ-2	Internal air temperature, Internal air relative humidity	15 min	Wind speed	Air temperature, Air relative humidity, Soil temperature, Soil moisture, Photosynthetically active radiation PAR, Pyranometer, Leaf wetness	35,040 samples	23/07/2020–22/07/2021	Private
[56] (2025)	Celery	Classical machine learning	RBF, MLP trained + Levenberg–Marquardt (LM)	30 min: Air temperature: RMSE = 1.579 °C, $R^2 = 0.958$ (RBF). 30 min: Air relative humidity: RMSE = 4.299%RH, $R^2 = 0.948$ (MLP). Current-time prediction: Air temperature RMSE = 0.439 °C, $R^2 = 0.997$ (MLP); Air relative humidity RMSE = 1.141%RH, $R^2 = 0.996$ (MLP)	RS-CO2WS-N01-2; RS-RA-N01-AL; KE-N01-TR-1; HOBO U30 NRC	Internal air temperature, Internal air relative humidity	10 min	Air temperature, Air relative humidity, Wind speed	Air relative humidity, Light intensity, Air temperature, Soil temperature, Soil water content, Light intensity	8441 samples	22/12/2021–19/02/2022	Public
[57] (2022)	NR	Deep sequential learning	DNNR, LSTM, CNN-LSTM	DNNR was the best overall model for 3 h, 6 h, and 24 h ahead prediction, while LSTM/CNN-LSTM performed better at 12 h ahead. Reported best-model error ranges: Air relative humidity (DNNR)—3 h: [−8, 10], 6 h: [−12, 13], 24 h: [−19, 26]; Air temperature (DNNR)—3 h: [−5, 8] °C, 6 h: [−6, 6] °C, 24 h: [−7, 7] °C. For 12 h ahead, LSTM/CNN-LSTM had smaller errors than DNNR	RS-WS-N01-2, RS-GH	Internal air temperature, Internal air relative humidity	1 min	Mean air temperature, Maximum air temperature, Minimum air temperature, Air relative humidity	Air temperature, Air relative humidity	1073 samples	GH 1: 1/05/2018–20/07/2018, GH 2: 01/05/2020–20/06/2020	On request
[58] (2025)	Cucumber, Melon	Hybrid modeling	Transformer + BiLSTM	MSE = 0.0267 °C, $R^2 = 0.9995$	Onset; Hukse-flux; Kipp & Zonen; Campbell Scientific	Internal air temperature	5 min	Air temperature, Air relative humidity, Solar radiation	Air temperature, Air relative humidity, Soil temperature at 30 cm	NR	GH 1: 01–03/2024, GH 2: 12–03/2024	Private
[59] (2024)	NR	Hybrid modeling	GCAKF-CNN-LSTM	Air temperature: RMSE = 0.065 °C, MAE = 0.032 °C, Air relative humidity: RMSE = 1.103%, MAE = 0.346%	NR	Internal air temperature, Internal air relative humidity	1 min	Air temperature prediction: Air temperature, Air pressure; Air relative humidity prediction: Air relative humidity, Air pressure	Air temperature prediction: Air relative humidity, air pressure; Air relative humidity prediction: Air temperature, Air pressure	25,497 samples	14/03–03/04 2019	NR
[60] (2025)	Tomato seedlings	Deep sequential learning	LSTM-AT-DP	Air temperature: $R^2 = 0.9602$, MAE = 1.4930 °C, RMSE = 1.6843 °C. Air relative humidity: $R^2 = 0.9529$, MAE = 3.1278%, RMSE = 3.9557%. Solar radiation: $R^2 = 0.9839$, MAE = 12.5898 W/m ² , RMSE = 19.5745 W/m ²	SHT41, ISL89013	Air temperature, Air relative humidity, Solar radiation	15 min	NR	Air temperature, Air relative humidity, solar radiation	NR	04–08/2024 06/03–26/06/2025	On request

Table A2. Cont.

Ref./Year	Crop	Category	Architecture	Performance	Sensors/Weather Station	Target	Interval	Outdoor Input Variables	Indoor Input Variables	Dataset	Period	Dataset Availability
[61] (2025)	NR	Hybrid modeling	BO-LSTM	Air temperature RMSE = 0.626 °C, $R^2 = 0.986$; Air relative humidity (absolute) RMSE = 0.309 g/m ³ , $R^2 = 0.987$	NR	Air temperature, Absolute humidity	5 min	Air temperature, Air relative humidity, CO ₂ concentration, Wind speed, Solar radiation, Sunshine index	Air temperature, Air relative humidity, Solar radiation, Water vapor pressure, Canopy saturated water vapor pressure, Net radiation from the crop canopy, Temperature of the cover layer&Net radiation from the crop canopy	NR	29/12/2021–10/04/2022 3–10 04/2022, 21–29 12/2022, 1–8 02/2023	Private
[62] (2024)	Chili pepper seedlings, tomato seedlings	Hybrid modeling	CNN-SE-LSTM	Air temperature: MAE = 0.540 °C, RMSE = 0.755, $R^2 = 0.940$, Air relative humidity: MAE = 0.936%, RMSE = 1.618, $R^2 = 0.951$, Solar radiation: MAE = 1.586 W/m ² , RMSE = 3.417, $R^2 = 0.936$	S10A greenhouse data	Air temperature, Air relative humidity, Solar radiation	10 min	Air temperature, Air relative humidity, Dew point, Precipitation, Surface atmospheric pressure, Wind direction, Wind speed, Solar radiation	Air temperature, Air relative humidity, Solar radiation, Quilts status, Ventilation status, Sprayer status	13,632 samples	10/04–30/08 2023	On request
[63] (2024)	Tomato	Hybrid modeling	LSTM2	$R^2 = 0.962$, MAPE = 3.216%, RMSE = 1.196 °C	HMP60, SQ-215, WindSonic 1405-PK-100, GMP343, CR300	Air temperature	10 min	Air temperature, Air relative humidity, PPFD, CO ₂ concentration, Wind speed, Wind direction	NR	127,623 samples	GH1: 6/11/2019–08/12/2021, GH2: 1/10/2021–31/01/2022	On request
[64] (2024)	NR	Deep sequential learning	GRU	Average $R^2 = 0.8811$, RMSE = 2.056 °C	GTPK-02-17; JNGW100; GL840	Air temperature	1 min	Air temperature, Air relative humidity, Solar radiation	NR	NR	07/10/2021–05/11/2021	Public

NR: Not reported.

Table A3. Summary of RGB-based Disease Detection in SGHs.

Ref./Year	Crop	Task	Architecture	Performance	Inference Time	Model Size	Hardware	Camera	Dataset	Dataset Availability	Annotation Tool
[65] (2020)	Cucumber leaf	classification	EfficientNet-B4 + Ranger	Acc: 97%	NR	201.65 M	GPU 4× NVIDIA GeForce RTX 2080 Ti (11 GB VRAM each)	NR	2816 images (resized 3000 × 3000) augmented to 20,259 (training 380 × 380)	private	NR
[66] (2023)	Sunflower, Dry bean, Field pea (leaves)	detection	EfficientNetB4 + Adam optimizer	Ac: 95.56%, Pr: 96.86%, R: 93.65%, Sp: 97.25%, F1: 95.18%, AUC-ROC: 96.92%	NR	NR	GPU: NVIDIA GeForce GTX 1080	NR	1764 images (resized to 224 × 224)	available upon request	NR
[67] (2020)	tomato leaf	detection	MobileNetv2-YOLOv3	F1: 93.24%, AP: 91.32%, IoU: 86.98%	GPU: 246 fps/16.9 ms, CPU: 22 fps/80.9 ms	28 MB	CPU Intel Core i7-9800X, GPU 2× NVIDIA GeForce GTX 1080 Ti (11 GB VRAM each), RAM 32 GB	NR	2385 images	unavailable	LabelImg

Table A3. Cont.

Ref./Year	Crop	Task	Architecture	Performance	Inference Time	Model Size	Hardware	Camera	Dataset	Dataset Availability	Annotation Tool
[68] (2021)	tomato leaf	detection	YOLO-Dense	mAP: 96.41%	20.28 ms	NR	CPU: Intel Core i7-9750H, GPU: NVIDIA GeForce RTX 2060, RAM: 16 GB	RGB camera	15,000 images (resized 544 × 544)	available upon request	LabelImg
[69] (2021)	tomato + cucumber (leaves)	classification	PRP-Net (ResNet18)	Acc: 98.26%, Pr: 92.60%, Sen: 93.60%, Sp: 99.01%	NR	921.99 MB	CPU: Intel Core i9-9820X, GPU: NVIDIA GeForce RTX 2080Ti (11 GB), RAM: 64 GB	RGB camera	4284 images (Resized to 448 × 448)	NR	NR
[70] (2022)	tomato leaf	detection	SE-YOLOv5	Acc: 91.07%, mAP@0.5: 94.10%, P: 86.75%, R: 92.19%	50.63 ms (19.75 fps @ 640 × 640)	42.796 MB	CPU: Intel Core i7-9700, GPU: NVIDIA GeForce RTX 2060 Super, RAM: 16 GB, Framework: PyTorch 1.7.0, OS: Windows 10, CUDA: 10.1, cuDNN: 7.6.5	Sony IMX363 rear camera on a MI 8 mobile phone	150 images (4032 × 3024), resized to 640 × 640 after augmentation to 1036 images	NR	LabelImg
[71] (2022)	Strawberry leaf	detection	DAC-YOLOv4	mAP@0.5: 72.7%, F1: 0.716, mp: 75.5%, R: 68.2%	43 FPS, 20 FPS	25 MB	Training: CPU: Intel Xeon E5-2620 v2 (6 cores, 2.10 GHz), GPU: NVIDIA GeForce RTX 2080Ti (11 GB Memory), RAM: 64 GB Deployment: Jetson Xavier NX: 6-core NVIDIA Carmel ARM V8.2 CPU, 384-core NVIDIA Volta GPU, 8 GB LPDDR4 memory Jetson Nano: ARM Cortex A57 CPU, 128-core Maxwell GPU, 4 GB LPDDR4 memory	OPPO smartphone	1023 images (initially 1040 × 780, resized to 416 × 416), 764 augmented to 6112 training images.	Public	LabelImg
[72] (2022)	Strawberry leaves, flowers and fruits	detection	Multi-scale Feature Fusion Faster R-CNN (ResNet-50 backbone + FPN + CBAM)	mAP: 92.18%, AP (per class): 90.69–94.39%	229 ms	91.28 MB	GPU: NVIDIA RTX 3060, CPU: AMD R7-5800H, RAM: 16 GB	Sony RGB camera	3600 images (2160 training, 720 validation augmented to 14,400)	NR	LabelImg
[73] (2023)	Strawberry (leaves, flowers, pedicels, and fruits)	detection	YOLO-GIC-C (Improved YOLOv5s + GhostConv + Involution + CARAFE + CBAM)	Pr: 93.3%, R: 90.3%, F1 ≈ 91.8%, mAP@0.5: 94.7%	92.6 FPS	NR	GPU: NVIDIA RTX 2060 super 8 G GPU, CPU: Intel i5-12400F 2.5 GHz	NR	Strawberry Disease Recognition 2246 images	Public	LabelImg
[74] (2020)	tomato leaf	severity estimation	U-Net + VGG16	MJ: (0.217–0.606), ME: 6.0% to 21.6%	NR	NR	NR	20MP RGB, natural light, gray BG	18,005 images (resized 256 × 256)	public (PlantVillage)	COCO Annotator
[75] (2021)	Strawberry (leaves, flowers, pedicels, and fruits)	segmentation	Mask R-CNN + ResNet101	mAP@0.5: 82.43%	NR	NR	GPU: Nvidia Titan XP	RGB camera + online sources	2500 images (resized to 419 × 419)	Public	Labelme
[76] (2024)	14 different crop	classification	down-scaled dense residual connection	Acc: 96.75%, P: 97.62%, R: 97.59%, F1: 97.58%	31.7 ms	NR	CPU Intel Core i7-11800H (11th Gen, 2.3 GHz), GPU NVIDIA GeForce RTX 3080 Laptop (16 GB), RAM 32 GB	NR	(1) Plant Village 54,306 images (resized 256 × 256), (2) Strawberry Disease Recognition 2500 images (resized 256 × 256)	Public	NR

Table A3. Cont.

Ref./Year	Crop	Task	Architecture	Performance	Inference Time	Model Size	Hardware	Camera	Dataset	Dataset Availability	Annotation Tool
[77] (2024)	Basil	detection	YOLOv8 vs. YOLOv9	YOLOv8: P 95.78%, R 96.43%, F1 0.96@0.544, mAP50 97.39, mAP50–95 87.22/YOLOv9: P 95.93%, R 96.64%, F1 0.96@0.536, mAP50 97.22, mAP50–95 88.38	NR	NR	GPU: NVIDIA Tesla T4 (24 GB VRAM); CPU: Intel Xeon @2.20 GHz; test device: Android mobile device	NR	4596 images	Public	NR
[78] (2024)	Tomato, Cucumber, Eggplant	detection	YOLOv8n-vegetable	mAP@0.5: 92.91%, mAP@0.5: 0.95: 57.48%, P: 92.72%, R: 87.73%	271.07 fps	9.07 MB	GPU: NVIDIA GeForce RTX 3090 Ti, 24 GB VRAM	NR	800 video sequences (40,000 keyframe images, Cropped and annotated to 28,000 images 640 × 640)	partly public/full access on request	NR
[79] (2025)	Barley seedlings	detection	YOLOv8-DDS	P: 98.9%, R: 96.6%, mAP@0.5: 98.5%	57.6 ms	4.6 MB	Training: GPU: RTX 2080 Ti (11 GB), CPU: Intel Xeon Platinum 8255C (12 vCPU) Deployment: Jetson Nano (edge device, TensorRT acceleration)	Hikvision industrial RGB cameras	1042 images (resized to 640 × 640, augmented to 4380)	Partial public/full access on request	LabelImg
[80] (2025)	Cabbage, Shanghai Green, Napa Cabbage, Brassica Napus	multi-class detection	IM-AlexNet	Pr: 88.61%, Rec: 81.46%, F1: 84.9%, mAP: 88.91%	NR	NR	GPU: NVIDIA GeForce RTX 3090 (24 GB); CPU: Intel Xeon Gold 6230 (20 cores); RAM: 128 GB	RGB camera	8000 images (720 × 1920 resized to 256 × 256)	private/upon request	Labelme
[81] (2025)	tomato, cucumber, pepper	detection	YOLO-vegetable (Improved YOLOv10n)	Pr: 95.8%, Rec: 94.9%, mAP@0.5 = 95.6%	18.6 ms/14.7 GFLOPs	3.8 M	CPU: Intel(R) Xeon(R) Gold 5418Y processor, GPU: Nvidia GeForce RTX 4090 (24 GB VRAM), RAM 32 GB	RGB images	15,000 images (resized to 640 × 640)	NR	NR
[82] (2025)	Pepper	detection	YOLO-Pepper (Based on YOLOv10n with AMSFE, DFPN, SDH, Inner-CIoU)	Pr: 94.69%, Rec: 93.87%, mAP@0.5: 94.26%	115.26 FPS	2.51 M params (5.15 MB)	GPU: Nvidia GeForce GTX 1080 Ti	RGB images: Surveillance videos	8046 images (1920 × 1080 resized to 640 × 640)	Public	LabelImg
[83] (2025)	Cucumber	lesions detection	YOLO-Cucumber (baseline YOLOv11n) + C3k2-DCN + P2 small-target layer (remove P5) + TAL + CPBN pruning	mAP@0.5: 93.8%; mAP@0.5:0.95: 72%; P: 93%; R: 92%	218 FPS	1.48 MB	GPU: NVIDIA GeForce RTX 3090 24 G; CPU: Intel Xeon Gold 5220; RAM: 64 G	Camera: Canon EOS 90D; lenses: 18–135 mm and 100 mm macro	4740 images (456 × 2304 pixels)	Partly public/full on request	LabelImg
[84] (2024)	Cucumber	small-target detection	CucumberDet (RetinaNet + Swin Transformer backbone + SFFM + MFAFM (and removes P7 layer))	mAP: 92.5%; P: 92.9%; R: 92.0%; F1: 92.4%	23.6 FPS (paper also reports 72 FPS in a separate comparison table)	38.39 M	GPU: NVIDIA GeForce RTX 3090 24 GB; CPU: Intel Core i9-10900X; RAM: 128 GB;	Not specified	4740 images	Public	LabelImg
[85] (2024)	Tomato	detection	TomatoDet (YOLOv8n + Swin-DDETR self-attention feature extraction module + Meta-ACON dynamic activation + IBiFPN multi-scale feature fusion)	mAP@0.5: 92.3%	46.6 FPS	43.9 MB; Params: 13.3 M	GPU (training): NVIDIA RTX 2080 Ti; GPU (speed test): Tesla T4 16 GB.	HS-CQAI-1080 agricultural IoT monitoring device; capture distance 0.2–0.5 m	2000 images (captured at 3648 × 2056) augmented to 9600 images	Partly public/full on request	LabelImg

Table A3. Cont.

Ref./Year	Crop	Task	Architecture	Performance	Inference Time	Model Size	Hardware	Camera	Dataset	Dataset Avail-ability	Annotation Tool
[86] (2025)	Tomato (leaf)	classification	TLDVLM: GroundingDINO (leaf detect)+SAM-2(segmentation)+BLIP-2 with LoRA on Q-Former	Acc 97.27%, P 0.9587, R 0.9789, F1 0.9681	2.7 B (base) + 0.6 M (LoRA + classifier)	NR	NR	RGB camera	NR	NR	NR
[87] (2021)	Strawberry	detection	PlantNet (ResNet152 pre-trained on Plant-CLEF/LifeCLEF2017) + FPN, 2-stage cas-cade	mAP: 91.65%	0.662 s	NR	PU: NVIDIA GeForce TitanXP, 64 GB RAM, test: Intel Xeon E5-2650	RGB camera	4725 images (train + val 3309 augmented 39,708)	NR	NR

NR: Not reported.

Table A4. Summary of Plant Growth Monitoring and Yield Estimation studies in SGHs.

Ref./Year	Crop	Task	Growth Stage	Sensors	Data Type	Inputs	Interval	Dataset	Study Period	Model Family	Architecture	Data Split Strategy	Performance	Prediction Horizon	Dataset Availabil-ity
[96] (2025)	Tomato Normal/ cherry	Fruit ripeness/ object de-tection + classifica-tion	green, half-ripened, fully-ripened	DHT-11, LDR, SHT-10, Pi Camera	RGB images, IoT sensor	RGB images, Temperatures, Humidity, Light intensity, Soil moisture	NR	804 images (3024 × 4032, 3120 × 4160 resized to 640 × 640)	NR	deep learning	YOLOv8	train/test split: 643/161	mAP: 52.8% (avg re-call 0.478), Test accu-racy: 52%, Per-class: b = cherry l = normal b_fully_ripened: Pr = 0.444, R = 0.344, F1 = 0.388, Supprot = 93 b_half_ripened: Pr = 0.362, R = 0.313, F1 = 0.335, Supprot = 134 b_green: Pr = 0.625, R = 0.656, F1 = 0.640, Supprot = 369 l_fully_ripened: Pr = 0.465, R = 0.433, F1 = 0.448, Supprot = 289 l_half_ripened: Pr = 0.453, R = 0.419, F1 = 0.435, Supprot = 241 l_green: Pr = 0.489, R = 0.708, F1 = 0.578, Supprot = 638	NR	On request

Table A4. Cont.

Ref./Year	Crop	Task	Growth Stage	Sensors	Data Type	Inputs	Interval	Dataset	Study Period	Model Family	Architecture	Data Split Strategy	Performance	Prediction Horizon	Dataset Availability
[97] (2024)	Tomato	Instance segmentation + maturity classification + Correcting missed fruit counts +harvest readiness	Fruit detection(maturity classes): Green, Half-ripened, Fully ripened Plant condition(growth stage): 0 = Before ripening, 1 = Ripe in 1–3 weeks, 2 = Ripe within 1 week, 3 = Fully ripened, 4 = No detection	Logicoool C922n Pro webcam + CMUcam5 Pixy2	RGB images	RGB images	NR	Training: 392 images	10/2021–06/2022	CNN + Bayesian network	YOLACT + DBSCAN + Bayesian network	NR	Fruit detection: Overall F1 = 0.84	NR	Public
[98] (2022)	Mini-tomato (Milla)	growth monitoring	Flowering/fruit appearance→full ripening (rapid: day 5–25; slow: >25)	Raspberry Pi Zero W + 5MP camera + LED flash	RGB images	RGB images (cropped 1024 × 1024), binary masks	15 min (00:00–18:00); modeling: 4 imgs/day (03:00, 05:30, 12:00, 16:30)	11,232 captured; MTIL: 385 labeled images + masks	18/07/2019–28/10/2019	DL semantic segmentation + nonlinear regression	TommyNET (modified symmetric U-Net w/multi-scale residual blocks) + Gompertz	Train/Test/Val = 277/69/39 (by daytime) + 10-fold CV	Segm best: Precision = 0.99, Recall = 0.97, IoU = 0.95; Growth fit: $R^2 = 0.97$, sMAPE = 10.1%	NR	Public
[99] (2023)	Mushroom	Growth-stage detection, Growth monitoring (size + harvest alert)	- Stage1/Stage2/Stage3 (Stage3 = ready to harvest) - Mask pixel area → growth rate + harvest time	Stage1 = first days/very small; Stage2 = formed, growing (not ready); Stage3 = cap edges flatten/slightly up-rolled (ready)	RGB camera	RGB images (640 pixels)	NR	1128 labeled imgs, 4271 instances (S1 = 1130, S2 = 1845, S3 = 1296); +231 photo monitoring (33 bags × 7 days); Detectron2 train sets: 453 mushrooms + 200 bags imgs	16/08/2022–24/08/2022	DL object detection + instance segmentation	YOLOv5s/ YOLOv5l (default + hyperparameter evolution); Detectron2 (2 models: bag det + mushroom instance masks)	YOLOv5: train/val = 784/344; Detectron2-mushrooms: 358/95; Detectron2-bags: 150/50	YOLOv5 (v5l + evolved HP + bs = 8): mAP@0.5 = 0.79353; mAP@0.5:0.95 = 0.54582, Precision = 77.89%, Recall = 75.44%, F1 = 76.65%, Per-class acc: Stage1 = 82%, Stage2 = 71%, Stage3 = 70%, Monitoring: avg growth period = 5.22 d; harvest earlier for 17 mushrooms (=14.04%); size threshold = 5.34%	NR	On request
[100] (2021)	Tomato	Yield forecasting	NR	NR	daily time-series	recorded yield, CO ₂ , Temperature, Humidity deficit, RH, radiation	Daily; sliding window = 7 days, step = 1 day	GH1 (2018), GH2 (2017), GH2 (2018)	2017–2018	Deep learning (sequence)	LSTM+ TCN+ FC	70/30 train/test (per dataset)	RMSE (g/m ²): GH1: 10.45 ± 0.94; GH2: 6.76 ± 0.45; GH3: 7.40 ± 1.88	1 week	On request
[101] (2023)	Tomato	Yield Prediction	Full season	NR	time-series	recorded yield, CO ₂ , temperature, humidity deficit, RH, radiation	NR	2-year UK grower	NR	Biophysical + Deep learning + Fusion ensemble	Reduced TOMGRO + CNN-RNN + fusion (linear/Bayes/NN/RFR/GBR)	Year1 = train, Year2 = test	RMSE = 17.69 ± 3.47 g/m ² , $R^2 = 0.9995 \pm 0.0002$, NSE = 0.9989 ± 0.0004, PBIAS = 0.1791 ± 0.6837	NR	On request

Table A4. Cont.

Ref./Year	Crop	Task	Growth Stage	Sensors	Data Type	Inputs	Interval	Dataset	Study Period	Model Family	Architecture	Data Split Strategy	Performance	Prediction Horizon	Dataset Availability
[102] (2025)	cabbage, lettuce, spinach	crop growth multi-step prediction(Plant height, plant width)	NR	NR	Multivariate time-series	air humidity, air temperature, light intensity, soil moisture content, soil temperature, soil electrical conductivity, pH value, Nitrogen content, Phosphorus content, Potassium content	Env: 1 min; Growth-state: 3 × /day (09:00, 13:00, 17:00)	cabbage: 03/11/2023–25/11/2023, lettuce: 19/11/2023–13/12/2023, spinach: 08/12/2023–31/12/2023)	11/2023–01/2024	Hybrid ML+DL	SVR_Seq2Seq: SVR (RBF kernel) + Seq2Seq (Bi-LSTM encoder + LSTM decoder) + linear fusion + FC (output layer)	train 80%, test 20%	Height (cm): Next day (k = 3): MAE = 0.1239, RMSE = 0.1357, MAPE = 0.0123 2 days ahead (k = 6): MAE = 0.1500, RMSE = 0.1539, MAPE = 0.0147 3 days ahead (k = 9): MAE = 0.1881, RMSE = 0.1973, MAPE = 0.0185 4 days ahead (k = 12): MAE = 0.2242, RMSE = 0.2503, MAPE = 0.0226 Width (cm): Next day (k = 3): MAE = 0.3139, RMSE = 0.3281, MAPE = 0.0209 2 days ahead (k = 6): MAE = 0.3794, RMSE = 0.4063, MAPE = 0.0255 3 days ahead (k = 9): MAE = 0.4535, RMSE = 0.5237, MAPE = 0.0315 4 days ahead (k = 12): MAE = 0.4918, RMSE = 0.5347, MAPE = 0.0347	1 to 4 days	On request
[103] (2021)	sweet pepper	Fruit detection, fruit growth prediction	IM1: 35 days before harvest IM2: 25–35 days before harvest IM3: 15–25 days before harvest IM4: 15 days before harvest B (Breaker): 7 days before harvest M (Mature): harvest stage	RGB images, env time-series	NR	RGB images, env + growth	NR	7682 images (resized 1024 × 1024) 35,958 fruits labeled	26/02/2020–07/07/2020, 06/04/2020–24/06/2020	DL detection (CNN) + DL tabular classifier (MLP) + ensemble	Yolov5, MLP	80% train (then train/val = 70/30), 20% test	Ensemble (6 stages): mean F1 = 0.77, IoU = 0.86; CNN-only means F1 = 0.56; MLP-only means F1 = 0.49; CNN (3 stages in ensemble): F1 = 0.91, IoU = 0.86	7–35 days before harvest	NR

NR: Not reported.

Table A5. Summary of vision-based robotic harvesting systems in SGHs.

Ref./Year	Crop	Robot Platform	Architecture	Performance	Sensors	Perception Task	Success Rate	Cycle Time	Damage Rate	Dataset	Dataset Availability
[104] (2025)	Strawberry	NR	DHN-YOLO (YOLO11n-pose + CDC(CGA+DCNv3) + C3H(HetConv) + New-Neck	Fruit detection: P = 87.3%, R = 88.0%, mAP50 = 91.8% mAP50:95 = 78.6%; KeyPoint detection: P = 83.0%, R = 87.5%, mAP50 = 89.7%, mAP50:95 = 83%	Smartphone rear camera	Maturity-stage detection (unripe/ripe/fully ripe) + key picking point detection	NR	NR	NR	2066 images (1024 × 768 augmented to 5018 images)	upon request
[105] (2025)	Tomato	Vision-based detection + 3D localization module for a tomato picking robot	detection/YOLOv5 + stereo depth/3D position/SGBM; training uses SGD + Mosaic + warmup + cosine annealing	NR	Lena CV USB3.0 binocular camera (stereo)	Tomato detection in 3 classes: unoccluded/leaf-occluded/branch-occluded + 3D localization (depth) using stereo matching	Detection: 94.1% accuracy (YOLOv5; 95/102 correct) and mAP@0.5 = 0.947 (class mAPs: 0.941/0.953/0.946)	Localization processing time 1.16–1.51 s (avg 1.27 s)	NR	640 greenhouse images collected (0.7 m camera distance) augmented to 7788 images	On request
[106] (2024)	Sweet pepper	Universal Robot UR3 (6-DOF) controlled with MoveIt + RRT-Connect, ROS Noetic modules	Real-time semantic segmentation NN (MobileNetV2 bottleneck blocks + pyramid pooling; depth explored); 3D recon uses ORB-SLAM3 + ICP registration	NR	NR	Intel RealSense L515 RGB-D camera	Semantic segmentation of plant parts (stem/leaf/petiole/fruit/others) → 3D semantic point cloud → detect pruning position + pruning direction (end-effector pose)	57% success over 30 attempts (43% failures with reasons reported)	Harvest/prune cycle time NR; but 3D semantic point cloud creation = 0.2–0.3 s per 640 × 480 RGB-D frame	1000 sweet-pepper images	On request
[107] (2021)	cherry tomato	Mobile robot platform + robotic arm; eye-in-hand RGB-D camera. Workflow: detect bunch at 0.5–0.7 m, then arm moves camera to 0.3 m close-up for peduncle segmentation and pose estimation	Stage-1: YOLOv4-Tiny for tomato-bunch detection (coarse), Stage-2: YOLACT++ instance segmentation (R-101-FPN chosen) to segment bunch & peduncle masks, then least-squares curve fit + 3 keypoints + geometric model for pose	vision results: YOLOv4-Tiny Precision 92.7%, Recall 96.2%, F1 94.4%, speed 9.1 ms/img, YOLACT++ (R-101-FPN): mAP 73.1, mAP50 96.9, mAP75 87.3, 9.2 FPS (0.109 s/frame)	Intel RealSense D435i & D415 (RGB-D)	Peduncle cutting-point localization + peduncle pose estimation (yaw/pitch) using 2-stage detection/segmentation and geometric fitting	Peduncle keypoints found in 112/120 close-up tests (93.3%), Pose error (30 tests): yaw 4.98°, pitch 4.75° avg	NR	NR	1528 images total = 828 long-distance (0.5–0.7 m) + 700 close-up (0.25–0.35 m)	NR
[108] (2024)	marigold, snapdragon, pansy/horned pansy	Straddle cart over greenhouse benches (120 cm wide, 175 cm high) with Jetson TX2 for edge processing	Detection: YOLOv5l fine-tuned (FLOLO) + Segmentation: SAM (ViT-Huge); Pose: PCA on segmented point cloud; Plucking point: linear regression	NR	Stereolabs ZED 2/ZED 2i RGB-D stereo cameras	Ripe flower detection (2D), segmentation, 3D localization, pose estimation, plucking-point estimation	Detection (FLOLO test): mAP@0.5 = 0.68, P = 0.67, R = 0.68	NR	NR	FloraDet: 134 images/2443 flowers	Upon on request
[109] (2024)	Strawberry	Wire-driven multi-joint arm with spherical joints (3D printed), mounted on lifting module; module moves along aluminum extrusion track on hydroponic shelf; deployed on Jetson Nano; Arduino-based actuation mentioned	YOLOv4 detector (CSPDarknet53 + SPPNet + PANNet)	Detection: precision 0.891, recall 0.880, AP@0.5 = 0.891, AP@0.5:0.95 = 0.718 (avg over test periods), 90% detection accuracy	2× RGB cameras (Logitech Brio Ultra 4K HD)	NR	Up to 82% fruit picking success rate (FPR); as low as 32% early cloudy morning	Strawberry detection + 2D localization (center) + contour area estimation (used for depth/positioning); identify optimal picking points	NR	2000 RGB images (416 × 416 pixels)	NR

Table A5. Cont.

Ref./Year	Crop	Robot Platform	Architecture	Performance	Sensors	Perception Task	Success Rate	Cycle Time	Damage Rate	Dataset	Dataset Availability
[110] (2024)	Tomato	six-degree-of-freedom&Tracked mobile chassis, 5-degree-of-freedom robotic arm, lifting mechanism, collection basket, and laser radar	YOLOv5, HSV +fusion for mature-red extraction; nighttime denoising uses Gaussian + guided filtering	NR	ZED 2i stereo camera + laser radar RS-Helios-16P	RGB, depth map, 3D point cloud (stereo)	Day: 87.78%; Night: 87.55%	Day: 15.99 s/tomato; Night: 17.26 s/tomato	NR	Self-collected images: 8 images per tomato cluster from multiple angles (10°, 45°, 90°, 135°, 170°, plus −45°, 0°, 45°); scenes include single/multiple tomatoes with/without occlusion;	NR
[111] (2024)	Tomato	Rail cart system with vertical/horizontal mast + 2 fixed cameras (30 cm apart): Perpendicular camera (PC) + Angled camera (AC); an extra eye-in-hand camera dataset collected from a robot arm	Mask R-CNN (Detectron2)	NR	Intel RealSense D435 cameras	Instance segmentation/detection of tomatoes + stem parts	NR	NR	NR	Pose-PC: 100 labeled images (from 100 image pairs), 1080 × 1920, perpendicular view. Pose-AC: 100 labeled images, 1080 × 1920, angled view (25°). Illumination: 175 images, 2048 × 2448, collected in 2017 with poor illumination/overcast. Task: 184 labeled images, RealSense on robot arm, angled 54°, wider perspective	On request
NR: Not reported.											

Table A6. Overview of AI and IoT implementations in smart greenhouses.

Ref./Year	Crop	AI Task/Categorie	Architecture	Performance	IoT Communication Technology	Sensors	Data Type	Inputs	Target	Processing Level	Dataset	Periode	Dataset Availability
[139] (2022)	Olive	Irrigation, Fertilization decision/Classification	adaptive PSO-ANN	Acc = 94.8%, Pr = 91.15%, R = 97.93%, F1 = 94.42%, MAE 3.91	LM35, hygrometer, OMC-118, DHT22	time-series	NR	Olive type, temperature, soil moisture, wind, humidity	irrigation and fertilization	local control	MNIST, NSL-KDD, Syngenta Crop Challenge 2017, University of Arkansas plant dataset	NR	NR
[140] (2020)	Tomato	Frost forecasting(air temperature)+control/Hybrid	MLP-ANN (BP training)	effectiveness > 90%, R ² = 89.27–95.22%	XBee-WiFi; GSM/GPRS (M2M); TCP/IP (via WAP)	Weather station (SparkFun Weather Shield DEV-12081) + wind/rain sensors (external)	time-series	ANN: Outside air temperature, Outside air relative humidity, Wind speed, Global solar radiation flux, Inside air RH Fuzzy: Predicted ANN temperature, Cropland temperature	ANN: inside air temperature Control target: pump activation (%PWM)	NR	Edge + Web	1 year	NR
[141] (2022)	Gerbera, Broccoli	Greenhouse control	SVM regressor, MLP (ANN) regressor	RMSE (avg): SVM = 0.11, MLP = 0.08; reported accuracy = 92%	Wi-Fi + MQTT (TCP/IP publish-subscribe) to Adafruit IO Cloud; serial to PC	DHT11, LDR, MQ2	On/off time duration of pump, ventilation fan, amount of light	1024 samples per sensing parameter; 70/30 train-test split	10 days	NR	NR	NR	NR

Table A6. Cont.

Ref./Year	Crop	AI Task/Categorie	Architecture	Performance	IoT Communication Technology	Sensors	Data Type	Inputs	Target	Processing Level	Dataset	Periode	Dataset Availability
[142] (2023)	Greenhouse vegetables (tomato, onion, peas, etc.)	Greenhouse monitoring	DCNN (Dilated CNN) + Fire Hawk Optimizer (FHO)	Accuracy: 95%;	NR	DHT11, YL69, LDR/light sensor, smoke sensor, water level sensor (with fan/pump/bulb/LCD)	NR	Temp, humidity, soil moisture, light, smoke, water level	Normal vs. out-of-range condition + control actions (fan/pump/light/alarm)	Edge (Raspberry Pi) + Cloud	NR	07/–08/2022	NR
[143] (2022)	Tomato	Disease detection/classification + Fruit ripeness monitoring (instance segmentation/detection)	CNN; Mask R-CNN	Avg accuracy = 0.91; weighted = 0.93	IP network (IP camera); Arduino–Raspberry Pi via USB serial; socket-based transfer to server	DHT11, moisture sensor, water-flow sensor, IP camera	Time-series + RGB images	Leaf/fruit images (256 × 256); sensor readings	Leaf disease class (10 diseases + healthy) + fruit ripeness stage (3 classes)	Edge (Raspberry Pi)/Server	PlantVillage tomato diseases: 8500 imgs; fruit stages: 1500 imgs (10,000 total)	NR	PlantVillage public
[144] (2022)	Tomato	Flower/fruit monitoring, harvest prediction	Yolov3 + Prophet	High mAP/F1; harvest prediction error ± 2.03 days	Wi-Fi	5Camera (ESP32 + OV2640) + temperature sensor (Agrilog ITKOB0-Z)	RGB images + temperature time-series	RGB images + temperature	Fully bloomed flowers + immature fruits + harvest date	Edge (Raspberry Pi)	1350 labeled images (2 classes) + temp data (02/2019–01/2021)	NR	NR
[145] (2022)	Watermelon + Pumpkin seedlings (greenhouse)	growth-point detection + height estimation	EfficientNet (BiFPN)	Growth-point detection: AP = 96.6%, F1 = 94%, time = 0.026 s; Height: R ² = 0.92–0.97, RMSE = 2.81–4.83 mm	4G cellular module (upload to cloud); TCP/IP video stream; Wi-Fi hotspot (PC ↔ Raspberry Pi)	Intel RealSense D415 + Azure Kinect + surveillance camera; BH1750, SHT30, CCS811; Silan A1 LiDAR; 9-axis IMU	RGB-D images + env time-series + video	RGB-D seedling images + light, Temperature, RH, CO ₂	Seedling height	Edge (Raspberry Pi/STM32) + Cloud server	Labeled growth-point images: 1600 (after augmentation), split 90/10 train/test; test set mentioned: 160 images	Greenhouse test: 24/05/2022–26/05/2022	On request
[146] (2020)	Micro-tomato	Prediction/Regression	DBN (Dynamic Bayesian Network + EM)	Leaf Area Index (LAI): avg error 15.5%/12.2%/19.7% (Env1/2/3) and forecasts up to 21 days ahead; Evapotranspiration (ET) avg error 29.42%	Netsens Wireless Unit, Scale/weighing balance, Planimeter: DT Area Meter MK2	Time-series	NR	Growing Degree Days (GDD) (from temperature), Solar irradiance (R)	Leaf Area Index (LAI) and Evapotranspiration (ET)	NR	COLTIV@MI mini-greenhouse	1/04/2019–31/05/2019	NR
[147] (2023)	Leafy vegetables	Crop growth management/ Growth-rate prediction	KNN (K-Nearest Neighbor)	Acc = 93% (Coconut fiber + NFT: 93.3%); weighted avg 93%	NR	NR	Time-series sensor data + images	pH, EC, water temp, ambient temp, air temp/humidity	Crop growth rate/yield condition (leafy vegetables)	Cloud storage + offline analysis	Data from University of Agricultural Sciences (GKVK), Bengaluru; 70/30 train-test split	NR	NR
[148] (2023)	Tomato	Disease detection/Classification	CNN	CNN: >90% overall; 92% validation accuracy, per-class precision/recall mostly 0.96–0.99 IoT: 99% successful transmissions; max packet loss 2% and 44% energy reduction vs. no-sleep transmission	ZigBee/IEEE 802.15.4 (XBee S2C mesh) for sensor nodes; Wi-Fi for camera nodes; cellular interface from gateway to cloud	AM2315, SHT-10, SEN-08942, AM2302, OV2640 camera (ESP32-CAM)	time-series + RGB images	RGB leaf image (size: 200 × 200) env: air temperature, air humidity, soil moisture, soil temperature	Tomato leaf health state (healthy vs. disease class; 10 tomato categories)	Edge/Cloud	tomato subset used: 16,012 images (12,810 train/3202 validation)	NR	PlantVillage: public, IoT data: upon on request

NR: Not reported.

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